DRAFT V0.1 · CIRCULATED FOR ADVERSARIAL AI REVIEW

From the Qn Universe to QnIR v0.1

A condensed, self-contained specification of the CDQN / Qn universe — constitution, numeric substrate, symbolic layers, domain projections, and governance — with a first, deliberately exposed candidate for the QnIR intermediate representation on the path toward QnLang, the governed coding language for AI agents.

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Draft status. Version v0.1 — circulated explicitly for adversarial AI review (Google Deep Review). This document is a *condensed, self-contained restatement* of the canonical Qn universe specifications of the `cdqn` repository, followed by a first, deliberately exposed candidate definition of the QnIR intermediate representation (Milestone 2, status PENDING). The narrative throughout is written so that a reviewer has maximal liberty to attack the design on the grounds of consistency, logic, physical grounding, and completeness. Numbered open questions (QO-1 … QO-15) in Section 10 are the response format requested from the reviewer; Section 9 pre-states anticipated objections so they can be sharpened rather than merely discovered.

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1 Introduction and Invitation to Attack

The Qn universe is a pre-silicon computational design corpus whose canonical documents [1–18] propose to reconstruct computation from a single operational conjecture:

> *A number system can abstract any computable phenomenon within a computational environment if, and only if, that abstraction is strictly governed by finite physical realities and constrained by the SIMEMP framework [1].*

Every artifact in this universe is a governed, finite, identifiable, measurable, and security-constrained computational entity [9]. The corpus is internally interconnected: a unified glossary binds every term, each specification cites its normative parents, and a single decision policy — the Best of Choices (BOC) policy (Sec. 2) — arbitrates all design conflicts. This paper condenses that entire corpus into one self-contained document and then performs the step the canon itself defers: the derivation of a first candidate for QnIR, the Quang Numbers Intermediate Representation.

1.1 Why this draft exists, and how it wants to be read

The canon is explicit about its own epistemic state. All specifications are *Constructive Working Hypotheses* pending Layer 0 execution receipts [1] §2.2; QnLang syntax and the QnIR instruction set are listed as open items [2] §13 and [9] §14 item 3. The development methodology is a recursive search loop — *Proposal* → *Construction* → *Evaluation* → *Selection* — in which failure to execute within finite bounds refutes the proposed abstraction and forces an iteration [1] §2.

This draft is an instrument of that loop, not a claim of completion. It is written under the author's explicit instruction that the reviewing intelligence attack the design: the goal of the exercise is to detect *what is good, but mostly what is bad and needs to be corrected*, so that the design can advance. Consequently:

- Sections 2–7 restate the canon faithfully but compactly, preserving exact numbers, contract identifiers, and receipt names so that consistency attacks have concrete material.
- Each major section closes with an *invited objection* box stating where the authors themselves suspect weakness.
- Section 8 derives QnIR v0.1 — a full draft instruction set, state model, memory model, encoding, and execution semantics — as the highest-risk, highest-value artifact to attack.
- Section 9 enumerates sixteen anticipated objections in the voice a hostile reviewer might use; the reader is invited to sharpen, reorder, or refute them.
- Section 10 poses fifteen numbered open questions (QO-1 … QO-15) with section pointers, so a point-by-point review can be mechanically integrated.

Nothing in Sections 2–7 redefines canon; all derived or proposed material is confined to Sections 8–11 and is marked as v0.1 candidate material. Per Axiom 13 [9], any accepted correction will produce a *new versioned artifact* (v0.2), never an in-place mutation of this document.

1.2 A caution on tone and claims

The author is not a formally trained mathematician publishing finished proofs; the corpus is an architectural thesis in the exploratory register of a research program. Where the canon states a property (e.g., "zero drift by construction"), this paper restates the claim as the canon makes it and flags, in the invited-objection boxes, the places where the claim depends on assumptions a

reviewer may reject (e.g., that thermodynamic metering can be made platform-invariant, or that exact arithmetic remains competitive). The reviewer should assume nothing in this document has been empirically validated on physical silicon: the entire stack is pre-execution by design.

2 The SIMEMP Constitutional Layer

The constitutional specification [1] governs every other document; where conflicts arise, `simemp.md` governs technically, `abstractionLayers.md` structurally, and `LICENSE.md` legally [2] §Normative References. Because the author has identified `simemp.md` as the root document of the whole design, this section reproduces its constraint system in full before anything is built on top of it.

2.1 Dependencies Determinism

The formal heart of the constitution is a conditional-entropy postulate on state transitions. A transition S_t of any computational entity is fully determined if, and only if, its conditional entropy given its complete set of declared causal dependencies $\mathcal{D}(S_t)$ vanishes:

$$H\big(S_t \mid \mathcal{D}(S_t)\big) = 0. \tag{1}$$

If any dependency is implicit, hidden, or unmeasured, the available dependency set $\mathcal{D}'$ is a strict subset of $\mathcal{D}$, yielding $H(S_t \mid \mathcal{D}') > 0$, which manifests operationally as non-deterministic state drift, runtime instability, or unmodeled attack surfaces [1] §1. Equation (1) is the single most load-bearing statement in the corpus: every later mechanism (receipts, hash-consing, gateways, anti-drift contracts) is an engineering measure to keep it satisfied.

2.2 Thermodynamic grounding

Computation is treated as a non-equilibrium thermodynamic process, not an abstract symbol game:

- **Landauer's bound.** Any irreversible manipulation or erasure of information dissipates work $W \geq k_B T \ln 2$; at the canonical ambient temperature $T = 300\,\mathrm{K}$ this is $W \geq k_B T \ln 2 \approx 2.87 \times 10^{-21}$ Joules per bit [1, 3, 12].
- **The Memory Wall.** Transporting data across physical memory hierarchies dissipates bus power $\mathcal{O}(CV^2 f)$ and introduces latency; the Metric invariant therefore requires explicit accounting for bandwidth and locality, not only asymptotic complexity [1] §4.1.
- **Open-system entropy balance.** Following the Prigogine formulation, a computing system is an open dissipative structure: $dS = dS_{\text{internal}} + dS_{\text{exchange}}$ with $dS_{\text{exchange}} < 0$; internal order is maintained by exporting entropy through bounded, signed receipts [1] §4.2.
- **The Semantic Gap.** Translation between high-level intent (QnLang) and physical execution (QnIR) is named a *primary vulnerability*; the constitution mandates bounded, verifiable intermediate representations — the very artifact this paper drafts [1] §4.1.

2.3 The two-tier constraint architecture

SIMEMP divides its constraints into a non-negotiable tier and a trade-off-able tier [1] §3:

- **Tier 1 — Existential Invariants.** *Identity* (distinctly known, collision-resistant, versionable, causally rooted); *Metric* (explicit finite measurement envelope for memory, compute,

precision, uncertainty); *Security* (state transitions, causal integrity, and execution limits verifiable against finite adversarial models). An abstraction violating any of these is discarded.

- **Tier 2 — Operational Agilities.** *Efficiency*, *Modularity*, *Portability*. These naturally conflict with one another.

The arbitration rule is stated as a golden rule: an Operational Agility may always be sacrificed to preserve an Existential Invariant, but an Existential Invariant must never be compromised for an Agility [1] §3.2.

2.4 The No-Implicit Rule and DCC profiles

The No-Implicit Rule states: *no implicit entity, behavior, assumption, default, interpretation, or convention may cross a SIMEMP Gateway; only explicit Qn objects may pass between layers* [2] §3.2. Its enforcement instrument is the DCC PROFILE — a machine-readable contract of **D**ependencies (capability contracts, commitments, prerequisites), **C**onstraints (hard ceilings on memory, steps, recursion depth, precision), and **C**apabilities (permitted operations, transformation rights, export authorizations) [1] §4.3. DCC profiles are abstract contracts realized through STRUCTURAL INDIRECTION: a vulnerable concrete primitive (e.g., a signature scheme) can be swapped for a patched module by emitting an explicit migration receipt, without fracturing the abstract contract or the causal lineage beneath it.

2.5 Totality by Budget and receipts

Divergent or infinite loops are rejected by construction. Every governed operation must terminate within its declared metric budget — a discipline named METRIC EXHAUSTION — resolving to an explicit receipt from a closed status set [2] §3.6:

`SUCCESS`, `FAILURE`, `NO_SOLUTION`, `TIMEOUT`, `BUDGET_EXHAUSTED`, `INCONCLUSIVE`, `QUARANTINED`, `PRECISION_INSUFFICIENT`, `EXPRESSION_TOO_COMPLEX`, `REJECTION_BY_GATEWAY`, `DIVISION_BY_ZERO_REJECTED`, `ORDERING_INDETERMINATE_AT_ZOOM`.

Silent non-termination is prohibited (Axiom 6, Sec. 4.2); the full named-receipt catalog with emitting contexts is tabulated in Appendix B.

2.6 Stochastic workflows and the candidate boundary

Generative and probabilistic processes are admitted only behind an explicit epistemic boundary [1] §6: stochastic engines propose candidate structures, candidates are formalized, measured against constraints, and either selected and sealed by a deterministic policy (emitting a signed receipt) or terminated with a defined status. Under Axiom 14 [9], unverified candidates cannot cross a SIMEMP Gateway. This is the corpus's answer to the neural-symbolic integration problem: generation is free, but crossing into governed state is always deterministic, metered, and receipted.

2.7 The BOC Policy

The BEST OF CHOICES policy is the formal decision framework for every design conflict in the corpus [1] §7:

$$\text{Selection} \;=\; \arg\max_{c \in \mathcal{C}} \big(\text{SIMEMP Compliance}(c)\big), \tag{2}$$

preferring, among implementation paths, the option that minimizes implicit state and guarantees invariant preservation even at higher operational overhead. The policy is deliberately qualitative

— "SIMEMP Compliance" is not given a numeric objective function in the canon — and the reviewer is invited to attack precisely this point (Sec. 9, Objection 1). In this draft, every QnIR v0.1 design choice in Sec. 8 is annotated with the BOC reasoning that produced it.

Invited objection

The constitution fuses three independent disciplines — information theory (Eq. (1)), thermodynamics (Landauer, Prigogine), and software architecture (DCC, gateways) — with no formal bridge between them. A reviewer may ask whether $H(S_t \mid \mathcal{D}(S_t)) = 0$ is a testable property of artifacts at all, or a design aspiration dressed as a theorem; whether receipts "export entropy" in any physical sense or merely record logs; and whether the two-tier hierarchy can be applied mechanically or requires exactly the human judgment it tries to eliminate.

3 The Numeric Substrate: Q(0) … Q(9) and Qm

3.1 Genesis: from physical noise to governed state

No number in the Qn universe is assumed; every constant must declare an unbroken causal lineage back to physical hardware noise [3] §1. A node onboards across the Layer 0 → Layer 1 SIMEMP Gateway through a six-stage state machine [3] §2:

1. **Raw entropy harvesting** from thermal noise, ring-oscillator jitter, or shot noise; the harvested bitstream must satisfy a baseline sample volume $N_{\text{sample}} \geq 4096$ bits.
2. **Statistical health tests** per NIST SP 800-90B [33]: repetition count, adaptive proportion, and a min-entropy threshold $H_{\min} \geq 0.85$ bits/bit.
3. **Cryptographic conditioning**: $\mathcal{S}_{\text{seed}} = \text{KDF}(\mathcal{B}_{\text{raw}} \,\|\, \text{Digest}(\text{HostContext}))$.
4. **Inner-Ring PQC key generation** under structural indirection (post-quantum signature schemes are swappable; the private root key never leaves local memory).
5. **Genesis object assembly** of the local origin Q(0) under the canonical nine-block Universal Envelope.
6. **Sealing and receipt emission**, locking Q(0) into an immutable `SEALED` lifecycle state and emitting the Node Genesis Receipt.

Genesis is strictly local: $\forall A \neq B \implies Q(0)_A \neq Q(0)_B$; no universal global zero is admitted [3] §3. The first unit Q(1) derives from Q(0) by the minimal work transition — the Abstract Compute Unit U, calibrated to Landauer dissipation at ambient temperature, i.e. $W(U) \geq k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J — and establishes the baseline metric quantum and the genesis axis d_1. Single digits Q(1)… $Q(9)$ are then constructed inductively along d_1 by the discrete Successor Morphism $\mathcal{S}(x) = x + Q(1)$, with $\text{Parent}(Q(k)) = \{Q(k-1), Q(1)\}$ and $H(Q(k) \mid Q(k-1), Q(1)) = 0$ [4] §2. Decimal glyphs $0 \ldots 9$ are interface labels only; internal representation is pure successor composition (*base neutrality*).

3.2 The single-digit anchor spectrum

Each single digit is a governed artifact carrying a specialized DCC contract (*Secondary DCC Source*), selected for its structural role rather than its decimal value [4] §3–4:

Two algebraic constants are anchored within Category A as exact Diophantine constraints, not infinite decimals: the golden ratio ϕ (from $Q(5)$), $\phi^2 - \phi - Q(1) = Q(0)$, and the minimal metric root $\sqrt{2}$ (from $Q(2)$), $D^2 - Q(2) = Q(0)$, whose residual r_D is preserved in the metric envelope across zoom scales [4] §5. When prime anchors enter multi-axial Gaussian rings, they

Table 1: The single-digit anchor spectrum with algebraic status [4] §3.

Artifact	Contract	Structural role	Status
$Q(2)$	`DCC_ANCHOR_Q2_PARITY_v1`	Parity generator, binary modulus, constructive even/odd partitions	Prime basis generator
$Q(3)$	`DCC_ANCHOR_Q3_SIMPLEX_v1`	Minimal 2D simplex; foundation of simplicial meshes for DEC	Prime basis generator
$Q(4)$	`DCC_ANCHOR_Q4_QUADRANT_v1`	First composite; quadrant source	$Q(2) \times Q(2)$
$Q(5)$	`DCC_ANCHOR_Q5_APERIODICITY_v1`	Aperiodicity generator; Penrose/quasicrystal order; prevents periodic sync locks in swarms	Prime basis generator
$Q(6)$	`DCC_ANCHOR_Q6_SYMMETRY_v1`	Minimal non-abelian gauge (S_3); smallest perfect number $1{+}2{+}3 = 1 \cdot 2 \cdot 3$	$Q(2) \times Q(3)$
$Q(7)$	`DCC_ANCHOR_Q7_PROJECTIVE_v1`	Cyclic modulus; Fano plane order $\mathrm{PG}(2,2)$; octonion imaginary basis count	Prime basis generator
$Q(8)$	`DCC_ANCHOR_Q8_OCTET_v1`	Silicon memory word (8-bit octet); Layer 0/1 bus segmentation	$Q(2)^3$
$Q(9)$	`DCC_ANCHOR_Q9_RADIX_v1`	Single-digit radix-10 ceiling; decimal normalization boundary	$Q(3)^2$

exhibit the three classical behaviors: ramification ($2 = -i(1+i)^2$), splitting ($5 = (2+i)(2-i)$), and inertia ($3, 7$ remain prime in $\mathbb{Z}[i]$); and the aperiodicity of $Q(5)$ is formalized through the 5D-to-2D cut-and-project method (a periodic $\mathbb{Z}^5$ lattice sliced at irrational slope ϕ yields a deterministic, non-repeating Penrose quasicrystal) [4] §4.2–4.3.

3.3 The Qm numeric leaf tuple

Every fully evaluated numeric artifact in Qm (Quang Mathematics) is an immutable discrete LEAF TUPLE [5] §3:

$$\text{Qn Numeric Entity} \;\equiv\; \langle \sigma,\; q_z,\; z,\; r_z,\; d \rangle, \tag{3}$$

where $\sigma \in \{\ominus, \odot, \oplus\}$ is a relational orientation (an oriented displacement relative to local Q(0) along axis d, governed by the reflection involution $\mathcal{I}_d$ with $\mathcal{I}_d^2 = \mathrm{id}$; signed zeros are prohibited — if $q_z = 0$ and $r_z = Q(0)$, σ collapses to $\odot$); $q_z \in \mathbb{N}$ is the unsigned metric magnitude (*ontologically primary*, because physical resources are non-negative); $z \in \mathbb{N}$ is the zoom level parameterizing the resolution quantum $\delta_z(d) = Q(1)_d / \kappa(z)$ with κ a strictly monotonic resolution function; r_z is the conserved residual with $0 \le |r_z| < \delta_z(d)$, never discarded and never silently truncated; and d is the dimensional axis, remainders on distinct axes never combining [5] §3.1–3.5.

Scale expansion is the telescoping cascade [5] §4:

$$r_z = \big(q_{z+1} \cdot \delta_{z+1}(d)\big) + r_{z+1}, \qquad \mathcal{I}_{\text{total}}(X_d) = \sum_{k=0}^{z} \big(q_k \cdot \delta_k(d)\big) + r_z. \tag{4}$$

Zero precision drift occurs at any intermediate scale. If $r_z = Q(0)$, the representation is algebraically exact (TERMINAL EXACTNESS); all further quotients vanish, further evaluation would burn budget on idempotent iterations (forbidden by the Metric invariant), and the gateway emits `RECEIPT_TERMINAL_EXACTNESS` with `ConsumedWork: 0` [5] §4.2.

3.4 The R-algebra: exact morphisms over tuples

Elementary arithmetic is defined constructively over the tuples of Eq. (3) [5] §5. Addition concatenates exact magnitudes over the common quantum δ_z; subtraction is addition composed with reflection, $A - B \equiv A + \mathcal{I}_d(B)$. Multiplication composes orientation parity ($\oplus \otimes \oplus = \oplus$, $\ominus \otimes \ominus = \oplus$, $\ominus \otimes \oplus = \ominus$, $\odot \otimes \sigma = \odot$) and multiplies exact unscaled magnitudes, re-resolving to canonical tuple form by Diophantine partition. Division is *not* a rational projection but a bounded constraint equation: given A, B, find C, R such that

$$A = (B \times C) + R, \qquad B \neq Q(0), \quad 0 \leq |R| < |B|, \quad \sigma_C = \sigma_A \otimes \sigma_B. \tag{5}$$

Division by origin is intercepted at the gateway, emitting `RECEIPT_DIVISION_BY_ZERO_REJECTED` (status `REJECTED`, code `ERR_DIV_ZERO`) [5] §5.3.

Classical Diophantine undecidability (Hilbert's Tenth Problem, DPRM) is neutralized structurally, not solved: every evaluation is confined to an explicit logarithmic height ceiling $h(u/v) = \max(\log|u|, \log|v|) \leq H_{\max}$ declared in the metric envelope; by Northcott's theorem the solution set of bounded height is finite [23]; and every evaluation either terminates exactly, reaches Terminal Exactness, or halts on budget exhaustion via `RECEIPT_BUDGET_EXHAUSTED` [5] §5.3, cf. [19].

Exponentiation is a bounded monoidal fold $A^B = \prod_{j=1}^{B} A$ with height growth $h(A^B) = B \cdot h(A) \leq H_{\max}$ (overflow emits `RECEIPT_EXPRESSION_TOO_COMPLEX`); negative exponents map to exact rational inversions $A^{-B} \equiv Q(1)/A^B$; 0^0 is indeterminate unless the calling profile declares `DCC_ZERO_EXPONENT_v1`; roots are polynomial Diophantine constraints $\sqrt[n]{A} \Rightarrow X^n - A = Q(0)$; an even root of a negative scalar cannot exist in 1D (no σ with $\sigma \otimes \sigma = \ominus$) and is promoted to a dimension-2 bivector blade $\vec{e}_1 \wedge \vec{e}_2$ in Clifford algebra; and the factorial is a fold with $\mathcal{O}(1)$ bypasses at 0!, 1!, and the fixed point 2! = 2, Q(reuse) admitted only for $n \geq 3$ (breakeven), and negative factorials structurally impossible because they would require division by Q(0) [5] §5.5–5.6.

Ordering is positional and bounded: comparisons evaluate constructive intervals $I(A) = [q_A\delta_z + r_A,\ q_A\delta_z + r_A + \epsilon_z]$; when intervals overlap and inequality cannot be decided at zoom z, heuristic tie-breaking is forbidden and the deterministic status `ORDERING_INDETERMINATE_AT_ZOOM` is emitted, requiring the caller to fund a zoom deepening $z \to z + \Delta z$ [5] §6.

3.5 The positive/negative asymmetry

The corpus records an asymmetry the reviewer may find unusual: the positive cone $\mathcal{Q}^+$ is a free commutative multiplicative monoid generated by the primes $\{Q(2), Q(3), Q(5), Q(7)\}$, while $\mathcal{Q}^-$ is *not closed* under multiplication ($\ominus \otimes \ominus = \oplus$ exits the negative domain). Dynamically, the Collatz map exhibits a single attractor cycle ($Q(4) \to Q(2) \to Q(1)$) on $\mathcal{Q}^+$ but fractures into at least three non-trivial limit cycles on $\mathcal{Q}^-$; thermodynamically, negative artifacts have identical positive physical footprints (executing $\ominus$ still dissipates $W \geq k_B T \ln 2 > 0$); and in quadratic integer rings the Dirichlet unit theorem expands the unit group beyond $\{\pm 1\}$ to infinite cyclic groups generated by fundamental units such as $\epsilon = 1 + \sqrt{2}$, giving discrete dilation symmetries without floating point [4] §7.

3.6 What Qm is not

The specification closes with an explicit demarcation table [5] §1: Qm is constructive discrete arithmetic, not IEEE 754; it conserves residuals, it does not silently truncate; signs are geometric involutions, not ontological entities; operations are total-by-budget, never infinite expansions; Diophantine relations are height-bounded, never unbounded Hilbert-10 void searches; and Qm is neither an authoring language (QnLang), nor an execution backend (QnIR), nor a symbolic

container (Qexpr) — exactly the boundaries that make a separate intermediate representation necessary.

Invited objection

The numeric layer's most attackable joints: (i) the resolution function $\kappa(z)$ is declared but never fixed — different choices yield different arithmetic, and the canon does not say who chooses; (ii) Terminal Exactness makes exactness decidable only *relative to a zoom budget*, so "exact arithmetic" is exact modulo an expense decision, which is a different epistemic claim than exact arithmetic over $\mathbb{Q}$; (iii) the Collatz discussion uses *tested* orbits rather than proved convergence, yet the corpus elsewhere rejects unmetered empirical testing; (iv) constructive parity and negative-domain dynamics are stated for single digits and asserted inductively, but no induction axiom over all naturals is ever given — the universe stops at $Q(9)$ and never formally constructs multi-digit integers, which the QnIR draft in Sec. 8 must inherit as an open gap (QO-4).

4 Layer Architecture, Qexpr CAE-DAG, and the Fourteen Axioms

4.1 The horizontal execution core

The physical execution stack terminates strictly at Layer 3, which lowers into the virtual execution handler (QnIR) [2] §2:

- **Layer 0 — Commodity Physical Substrate.** CPUs, GPUs, DSPs, FPGAs, memory hierarchies, storage, network interfaces; physical boundaries (finite memory, clock latency, thermal limits, Memory Wall); raw entropy sources. Layer 0 is the *Root of Finiteness* and remains outside the governed universe.
- **Layer 1 — Node Genesis Layer.** Q(0), Q(1), the compute unit U, sign polarity, single digits $Q(2) \dots Q(9)$ along d_1 (Sec. 3.1).
- **Layer 2 — Governed Morphism Layer.** The elementary $\mathcal{R}$-algebra, the optimization morphisms Q(reuse) and Q(bypass), inductive multi-axial frame extension $\mathcal{D}_{\text{step}}$, reflection involutions and Clifford geometric products, and secondary DCC routing to single-digit capability contracts.
- **Layer 3 — Symbolic Composition Layer (CAE-DAG).** Content-addressed expression graphs with hash-consing, structural equivalence in $\mathcal{O}(1)$, and deterministic collapse to canonical Qm leaf tuples or *flat linear instruction streams (QnIR)* upon crossing execution boundaries.

Inter-layer transitions are Universal Functorial Gateways $\mathcal{F} : \mathbf{C}(L) \to \mathbf{C}(L+1)$ that enforce the No-Implicit Rule and audit Landauer dissipation at every state step. Crucially, the domain projections (Qm, Qs, Qlog, Qphy, Qchem, Qbio) are *orthogonal vertical slices* across Layers 1–3, not horizontal layers above the stack; QnLang sits outside the core as a compilation frontend targeting Layer-3 CAE-DAGs; and distributed coordination (cdqn, Chronosa) operates transversally over emitted receipts, requiring no Layer 4 [2] §2.5.

4.2 The fourteen provisional operational axioms

The Qn operational model is governed by fourteen axioms [9] §3, reproduced here in condensed form because QnIR v0.1 will cite them individually as design requirements:

Axiom	Statement (condensed)
1 — Explicit Envelope	No artifact exists unless it carries an explicit Universal Envelope: $\forall \alpha \in \mathcal{Q},\ \exists! \mathcal{E}(\alpha)$.
2 — Finiteness	Every artifact has a finite canonical representation and bounded resource envelope; actual infinities are excluded from execution.
3 — Typed Payload Separation	Numeric fields (z, r, d) belong to typed payload profiles; they are not mandatory for non-numeric entities. Sovereign user data remains an uninspected Payload.
4 — Local Genesis Dependency	Every artifact traces its lineage directly or transitively to the local origin Q(0); higher-order network entities are topological colimits (sheaf gluings) over forests of local Axiom-4 trees.
5 — Acyclic Causal Lineage	Lineage graphs are strictly acyclic; Index(Parent(α)) $<$ Index(α).
6 — Totality of Governed Operations	Every operation terminates in a declared terminal state (the closed receipt set of Sec. 2.5); silent non-termination is prohibited.
7 — Receipted Boundary Events	Receipts are mandatory for lifecycle transitions, gateway crossings, sealing, export decisions, fault translations, and capability invocations.
8 — Local-first Exposure	Raw local artifacts are never exposed across the network without explicit authorization and bounded public attestation.
9 — Numeric Precision Explicitness	Every numeric artifact declares finite precision bounds, explicit z, r, d; implicit rounding and truncation are prohibited.
10 — No Floating-Point Semantics	IEEE 754 semantics are forbidden at the Qn semantic layer; non-deterministic mantissa rounding is rejected.
11 — Bounded Self-Description	Governance metadata may describe its own schema, but self-description depth must remain finite, bounded, and version-anchored.
12 — License Lineage Where Applicable	Lineage metadata and the paternity reference must be preserved in distributed artifacts, exported attestations, and compiled QnIR binaries.
13 — Versioning Creates New Artifacts	Versioning produces a new immutable artifact with an advanced causal index; mutation in place is prohibited.
14 — Generative Candidate Boundary	Unverified candidates (stochastic, neural, heuristic) can exist only when explicitly tagged; they cannot cross a gateway or achieve sealed execution without deterministic validation.

4.3 Qexpr: the Content-Addressed Expression DAG

Under the BOC policy, Layer-3 expressions are Content-Addressed Expression DAGs (CAE-DAGs), not pointer-heap trees; the rejection of pointer heaps is BOC-recorded and is one of the corpus's few explicit design verdicts [8] §2, [16] §4. Every symbolic node is an immutable tuple

$$\alpha = \left\langle \text{OpCode},\ \mathbf{D}_{\text{children}},\ z,\ r,\ d,\ \text{DCC}_{\text{ref}} \right\rangle, \qquad \mathbf{D}_{\text{children}} = [\mathcal{H}_1, \ldots, \mathcal{H}_k], \tag{6}$$

with the child commitment vector strictly arity-indexed (no implicit padding for nullary or unary operators), and node identity computed by

$$\mathcal{H}(\alpha) = \mathrm{Hash}\big(\mathrm{OpCode} \,\|\, k \,\|\, \mathcal{H}_1 \,\|\, \ldots \,\|\, \mathcal{H}_k \,\|\, z \,\|\, r \,\|\, d \,\|\, \mathrm{DCC}_{\mathrm{ref}}\big). \tag{7}$$

Consequences the canon draws: identical sub-expressions share one physical node (*hash-consing* deduplication); structural equality collapses to $A \equiv B \iff \mathcal{H}(A) = \mathcal{H}(B)$, an $\mathcal{O}(1)$ integer comparison; and acyclicity is guaranteed *by construction*, because a cycle would require a child hash to appear before the parent hash whose pre-image it is — breaking pre-image resistance [8] §2.2–2.4. Symbolic normalization inside the DAG is reversible and therefore claims zero Landauer dissipation ($W = 0$); memory traversal is billed, so sub-expression reuse is admitted only under the thermodynamic breakeven

$$\mathrm{Admit}\big(Q(\mathrm{reuse})\big) \iff \Delta W_{\mathrm{compute}} > \Delta W_{\mathrm{lookup}} + \epsilon_{\mathrm{overhead}}, \tag{8}$$

with strict dispatch precedence $Q(\mathrm{bypass}) \prec Q(\mathrm{reuse}) \prec$ Direct Compute [9] §5.9. Rounding drift is eliminated by holding expressions in exact symbolic form and partitioning into quotient and conserved remainder only at boundary demand (Eq. (4)); irrational and transcendental quantities exist *permanently* as symbolic generators — algebraic irrationals as constrained polynomials under `OP_ROOT`, transcendentals as deterministic continued-fraction generators under `OP_CF` (e.g. $\pi = [3; 7, 15, 1, 292, \ldots]$, $e = [2; 1, 2, 1, 1, 4, \ldots]$) [8] §3.

The six canonical node classes are: `OP_RATIO` (exact rational pairs, k=2), `OP_ROOT` (Diophantine radical polynomials, k=1), `OP_CF` (projective continued-fraction generators, k=0), `OP_FOLD` (bounded monoidal accumulation $\sum, \prod, n!$ with explicit step ceiling, $k \geq 3$), `OP_FRAME_MAP` (Clifford geometric products, rotor conjugations, coordinate permutations, k=2), and `OP_REWRITE` (equality saturation and the canonical $\mathcal{O}(1)$ algebraic identity bypass suite, $k \in \{1, 2\}$) [8] §4. The bypass suite short-circuits $A + Q(0) \to A$, $A \times Q(1) \to A$, $A \times Q(0) \to Q(0)$, $0! \to Q(1)$, $1! \to Q(1)$, $2! \to Q(2)$, $A^{Q(0)} \to Q(1)$, $A^{Q(1)} \to A$, $Q(1)^k \to Q(1)$, $Q(0)^k \to Q(0)$ $(k > 0)$, $\sqrt[n]{Q(1)} \to Q(1)$, rotor normalization $RR^\dagger \to Q(1)$, exterior nilpotency $\vec{e}_i \wedge \vec{e}_i \to Q(0)$, bivector inversion $(I_2)^2 \to -Q(1)$, and Poincaré coboundary $\mathbf{d} \circ \mathbf{d} \to Q(0)$ — each emitting `RECEIPT_BYPASS_IDENTITY` at $\Delta W = 0$ [8] §4.6.

4.4 HCA-Collapse: hardware-aware lowering

Physical execution is governed by the Hardware Capability-Aware Collapse Engine, which lowers Layer-3 DAGs along the node's local DCC hardware profile [8] §5: NPU/systolic targets receive integer Matrix-Multiply-Accumulate rewritings over INT16/INT32; cryptographic co-processors absorb hash-consing commitments (SHA-3, BLAKE3); and integer RISC-V targets use Barrett reciprocal tables with exact remainder-correction loops instead of division. Two invariants constrain all lowering: lowering must be an *algebraic isomorphism* (passes introducing floating-point approximation are rejected at the gateway), and lowered instruction sequences are constant-time so that execution duration depends strictly on declared metric budgets, providing hardware-level side-channel immunity [8] §5.2–5.3. Three structural circuit breakers bound fallibility for both human and machine authors: a static graph depth ceiling (`RECEIPT_EXPRESSION_TOO_COMPLEX`), a node-count budget (`RECEIPT_BUDGET_EXHAUSTED`), and a metric collapse budget guaranteeing deterministic halting under Axiom 6 [8] §6.

4.5 Relation to QnLang and QnIR in the canon

The primitive envelope specification fixes exactly three boundary statements about the execution substrate [9] §12: QnLang authors computations with explicit DCC profiles and prohibits implicit defaults and floating-point semantics; *QnIR flattens Content-Addressed Expression DAGs into linear, bounded instruction streams for deterministic hardware evaluation*; and

symbolic DAGs must collapse deterministically to canonical leaf tuples prior to terminal receipt emission. The virtual machine instruction set architecture and register schedules are explicitly excluded from qnPrimitive and named a Milestone-2 deliverable [9] §14 item 3, and `abstractionLayers.md` lists the "formal instruction set and operational semantics for QnIR" as open item 1 [2] §13. Section 8 of this paper drafts exactly that open item.

Invited objection

(i) The $\mathcal{O}(1)$-equivalence claim holds only for *syntactic* equality after canonicalization; any semantic equivalence still requires rewriting, and the canon's equality-saturation engine is asserted but never given termination or confluence guarantees. (ii) Hash-consing claims $W = 0$ for reversible normalization, but on physical hardware every comparison and table probe dissipates; the canon's own breakeven inequality (8) concedes this, and a reviewer may ask whether the $W = 0$ rhetoric is consistent with the Metered worldview. (iii) The axioms are called "provisional" and verified by "computational consistency", yet no theorem prover, model checker, or test suite is specified anywhere in the corpus; the verification loop is currently a document pattern, not a mechanism. (iv) Axiom 12 entangles a software license into the technical lineage invariant — an unusual coupling a reviewer may flag as category error or, alternatively, as a deliberate and defensible constitutional choice.

5 Geometry, Calculus, and Transcendental Anchors

5.1 Inductive dimensional genesis and Clifford frames

Multi-dimensional space is not assumed; it is generated. The genesis axis d_1 is instantiated by the transition Q(0) → Q(1) (basis vector $\vec{e}_1 \equiv \langle \oplus, 1, z{=}0, r{=}Q(0), d_1 \rangle$), and each further axis is adjoined by an explicit orthogonal step $\mathcal{D}_{\text{step}}$: $\vec{e}_{k+1} \cdot \vec{e}_i = Q(0)$ for all $i \leq k$, $\vec{e}_{k+1} \cdot \vec{e}_{k+1} = \eta_{k+1,k+1}\, Q(1)$, and $\text{Parent}(\vec{e}_{k+1}) = \{\vec{e}_k, Q(1)\}$, so every axis carries deterministic causal parentage back to Q(0) [6] §2.1. The quadratic form is an explicit discrete signature $\eta = (p, q, s)$ with $p+q+s = n$ (space-like, time-like, degenerate), covering Euclidean space $\mathcal{C}\ell_{n,0}$ and spacetime algebras $\mathcal{C}\ell_{1,3}/\mathcal{C}\ell_{3,1}$; spherical, hyperbolic, and conformal geometries are modeled as subspace constraints ($X \cdot X = \pm Q(1)$, null cone) with rotor morphisms, and curved manifolds as simplicial triangulations with Regge deficit angles [6] §2.2–2.3. In conformal frames the null basis vectors $\vec{e}_{\text{origin}}, \vec{e}_{\text{horizon}}$ are strictly decoupled from the scalar ring zero Q(0) — conflation of vector basis elements with ring scalars is prohibited [6] §2.3.

The geometric product $\vec{u}\vec{v} = \vec{u} \cdot \vec{v} + \vec{u} \wedge \vec{v}$ decomposes into an exact symmetric scalar contraction $\vec{u} \cdot \vec{v} = \sum_k \eta_{kk} u_k v_k$ (evaluated on the Qm R-algebra) and an anti-symmetric wedge $\vec{u} \wedge \vec{v} = \sum_{i<j} (u_i v_j - u_j v_i)(\vec{e}_i \wedge \vec{e}_j)$ with exact anti-commutativity and nilpotency. A general multivector carries the state tuple $M \equiv \langle \sigma_M, \alpha_{\mathcal{B}}, z, r_M, \mathcal{D}_{\text{frame}} \rangle$ over the 2^n blade basis, with per-blade conserved remainders; the unit pseudoscalar $I = \vec{e}_1 \wedge \cdots \wedge \vec{e}_n$ provides Hodge duality $M^* = MI^{-1}$ without matrix inversion [6] §3–4.

5.2 Float-free rotation and distance

Rotations avoid transcendental trigonometry entirely. A rotation is the composite of two discrete reflections (Cartan–Dieudonné), $R = \vec{b}\vec{a}$, applied as the sandwich $\vec{v}' = R\vec{v}R^\dagger$; for exact rational angles the rotor is parameterized by Diophantine Pythagorean pairs (m, n) via the rational Cayley form

$$R = \frac{Q(1) - B}{Q(1) + B} = \frac{(m^2 - n^2) + 2mn\,(\vec{e}_1 \wedge \vec{e}_2)}{m^2 + n^2}, \qquad RR^\dagger \equiv Q(1), \tag{9}$$

whose unit norm is an identity rather than a numerical property: the canon notes the rotor can be applied 10^9 consecutive times without normalization cycles [6] §5.2. For arbitrary externally

supplied angles, the continued-fraction expansion of $\tan(\theta/4)$ generates a rational rotor cascade with conserved angular remainder r_θ, exact when $r_\theta = Q(0)$ [6] §5.3. Distance is the Diophantine square constraint $D^2 = (q_D\delta_z)^2 + r_D$ with bounded remainder $0 \le |r_D| < (2q_D\delta_z^2 + \delta_z^2)$; exactness again means $r_D = Q(0)$ [6] §6.

Geometric workflows run a three-stage lifecycle: symbolic composition on the CAE-DAG ($W = 0$), reversible normalization annihilating nilpotencies and canceling inverse rotors ($W = 0$), and a single metered collapse onto zoom z at boundary demand — deferring dissipation so that k consecutive rotor transformations cost one collapse instead of k [6] §7. Complexity is honestly declared: geometric products scale $\mathcal{O}(2^n)$ in blade count and rotor sandwiches $\mathcal{O}(n \cdot 2^n)$; dimensional ceilings $n \le \mathrm{Dim}_{\max}$ (typically $n \le 4$ for spacetime kinematics, $n \le 16$ for Qs semantic frames) are declared in the metric envelope, and exceeding a declared budget halts deterministically with `RECEIPT_DIMENSIONAL_BUDGET_EXHAUSTED` [6] §8.

5.3 Discrete Exterior Calculus

Continuous limits are replaced by three constructive mechanisms [7] §2: explicit lattice refinement steps $z \to z{+}1$ (each burning metered budget), conserved residual telescoping (Eq. (4)), and Terminal Exactness early halting. Fields are *cochains* on oriented simplicial complexes — 0-forms (potentials, temperatures) on vertices, 1-forms (electric fields, velocities) on edges, 2-forms (magnetic fluxes, vorticity) on faces [7] §3. The discrete exterior derivative $\mathbf{d}$ is the coboundary operator $\langle \mathbf{d}\omega, \sigma_{k+1}\rangle = \langle \omega, \partial\sigma_{k+1}\rangle$ satisfying the exact nilpotent identity $\mathbf{d} \circ \mathbf{d} \equiv Q(0)$, anchored as `DCC_ANCHOR_DEC_D_v1`; the Hodge star $\star\omega = \sum_{\sigma_k} \frac{|\star\sigma_k|}{|\sigma_k|}\langle \omega, \sigma_k\rangle(\star\sigma_k)$ holds dual volume ratios as exact `OP_RATIO` commitments; the Laplacian is $\Delta = \mathbf{d}\delta + \delta\mathbf{d}$; and Stokes' theorem $\sum_K (\mathbf{d}\omega)(K) \equiv \sum_{\partial K} \omega(\partial K)$ is an exact identity with zero truncation error — conservation laws (mass, charge, vorticity, Maxwell constraints $\mathbf{d}F \equiv Q(0)$, $\mathbf{d}\star J \equiv Q(0)$) hold identically, not asymptotically [7] §3, [12] §3.3.

5.4 Transcendental anchors and time

The transcendental constants are nullary continued-fraction generator anchors inhabiting the Layer-3 container [7] §4: π (`DCC_ANCHOR_PI_v1`, $[3; 7, 15, 1, 292, \dots]$), $\tau = 2\pi$ (`DCC_ANCHOR_TAU_v1`), e (`DCC_ANCHOR_E_v1`, $[2; 1, 2, 1, 1, 4, \dots]$), and $\ln 2$ (`DCC_ANCHOR_LN2_v1`, $[0; 1, 2, 3, 1, 6, \dots]$), the last calibrating the Landauer accounting quantum itself. Partial convergents p_k/q_k carry the strict residual bound $0 \le |r_k| < 1/(q_k q_{k+1})$; evaluating a constant at zoom z yields an exact rational quotient plus conserved residual, with Terminal Exactness early halting where remainders collapse.

Time is the strictly monotonic causal arrow, not an unmeasured parameter. Every dynamical state carries the lineage tuple $\mathcal{L}(S_k) = \langle \mathbf{P}(S_k), \kappa(S_k), \tau_{\text{type}}, \mathcal{W}_{\text{cum}}(S_k), \Pi_{\text{receipts}}(S_k)\rangle$ (parents, causal index, type, cumulative Landauer work, receipt set); the aperiodicity anchor `DCC_ANCHOR_Q5_APERIODICITY_v1` guarantees $\mathcal{H}(S_i) \neq \mathcal{H}(S_j)$ for $i \neq j$, giving an irreversible internal arrow of time without external clocks; and verification of an N-step simulation requires only the Anchor-Cut chain — origin, K signed anchor receipts, terminal state — collapsing audit complexity from $\mathcal{O}(N)$ to $\mathcal{O}(K)$, $K \ll N$ [7] §5. Singularity immunity is structural: spatial gradients are bounded by the lattice quantum δ_z, viscosity is an explicit dissipative sink exporting entropy via receipts, and any incipient blowup halts at budget exhaustion before unphysical states can compile [7] §7.

Invited objection

(i) Rational Cayley rotors generate only rational-angle rotations exactly; arbitrary angles require the continued-fraction cascade, so "float-free" is exact only modulo an unbounded in principle convergence process — the same zoom-budget caveat as the numeric layer. (ii) $\mathcal{O}(2^n)$ blade growth means Clifford operations are practical only at the declared small dimensions; the canon's own budget halts concede this, but the corpus never states which applications survive the ceiling. (iii) DEC conservation is exact on the mesh, but the mesh itself is a modeling choice; the corpus does not state error bounds between the discrete solution and the modeled physical phenomenon (spatial discretization error is not rounding error, and defending the former is not provided by defending the latter). (iv) The Anchor-Cut invariant delegates trust to checkpoint receipts; the mechanism by which an anchor receipt is itself verified against a faulty-but-signed past (compromised anchor fracture is discussed in Qlog but not formally) remains open.

6 Domain Projections: Qs, Qlog, Qphy, Qchem, Qbio

The five domain specifications are orthogonal vertical projections across Layers 1–3. They are summarized here because a QnIR program must be able to carry their payload types and honor their gateway checks; each subsection ends with the domain's anchor spectrum (full registry in Appendix C).

6.1 Qs — Quang Semantics

Qs formalizes language, meaning, and intent as constructive, multi-axial relational structure [10]: language is a structured graph of discrete semantic blades, not a black-box tensor; meaning is an explicit, verifiable Intent Contract bound to human authority, not statistical similarity. Vocabulary is modeled in Clifford geometric algebra over semantic dimensions $\mathcal{D}_{\text{semantic}} = \{d_{\text{intent}}, d_{\text{consent}}, d_{\text{obligation}}, d_{\text{temporal}}, d_{\text{agent}}, \dots\}$: nouns are grade-1 vector blades $\vec{v}_{\text{noun}} = \sum_k \langle \sigma_k, q_{z,k}, z, r_{z,k}, d_k \rangle \vec{e}_k$; modifiers are rational Cayley rotors $\vec{v}' = R\vec{v}_{\text{noun}}R^\dagger$ with $R = (Q(1) - B)/(Q(1) + B)$, so modification preserves unitary norm by construction; and transitive verbs are conserved relational morphisms Action = (Delegation × Actor) + r_{social}, where the conserved social residual r_{social} keeps unfulfilled obligations permanently visible in the lineage — silent state abandonment is impossible [10] §3.

The EPISTEMIC ANCHORAGE SPECTRUM is Qs's anti-hallucination firewall; every statement must declare its modality [10] §4:

Table 3: The epistemic anchorage spectrum [10] §4.

Modality	Grounding	Semantics
`MODAL_EMPIRICAL`	$G = 1.0$	Anchored to Layer-0 physical sensor receipts through a SIMEMP Gateway.
`MODAL_CONSTRUCTIVE`	$G = 1.0$	Algebraic closure proven via Qm/DEC with remainder $r = Q(0)$; cannot drift by construction.
`MODAL_HYPOTHESIS`	$0.3 < G < 1.0$	Explicitly bounded working conjecture inside the Recursive Search Loop, awaiting physical falsification.
`MODAL_NARRATIVE`	contextual	Internally consistent fiction anchored to author identity $Q(0)_{\text{author}}$; barred by contextual containment from triggering actuators, financial registers, or legal delegations.
`MODAL_UNANCHORED`	$G = \emptyset$	Claims truth without parent lineage receipts; *rejected at the gateway* with `RECEIPT_UNANCHORED_ASSERTION_REJECTED`.

Alignment between humans and agents is enforced structurally, not by fine-tuning: the INTENT CONTRACT (`DCC_INTENT_CONTRACT_v1`) embedded in a DCC profile declares the sovereign

human anchor, the agent proxy, the permitted multi-axial domain, the maximum work allocation W_{allowed} in Q(1) units, the allowed epistemic modalities, and an immediate revocation path [10] §5. Under the Non-Rogue Invariant an AI agent has zero independent root sovereignty: it cannot instantiate Q(0), executes only under an unbroken delegation chain from a human principal, and halts physically when its work budget reaches zero (the *thermodynamic leash*). Communication with generative models is routed through the Q(rsi) recursive loop: candidates proposed by neural engines are parsed into CAE-DAGs, measured against the intent contract, selected by BOC (Eq. (2)), and sealed as successor artifacts with advanced causal indices [10] §6. A command executes only under TRIADIC GROUNDING $\mathbf{C}_{\text{Qs}} \otimes \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qphy}}$: intent (is the human authorized), decidability (is the trajectory algebraically exact), and realizability (does the battery and heat budget permit it) must all verify before any actuator moves [10] §7.

6.2 Qlog — Quang Logics

Qlog is the composite domain $\mathbf{C}_{\text{Qlog}} = \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qs}}$ [11]. It rejects non-constructive deduction: the Law of Excluded Middle, unmetered search in infinite voids, and ephemeral textual proofs are discarded in favor of VERIFIED COMPUTED PATTERNS — content-addressed state trajectories sealed by immutable receipts with $H(S_{\text{out}} \mid \mathcal{D}(S_{\text{out}})) = 0$ [11] §1. Two causal entity classes organize execution: a Q(PATTERN) is an invariant state-transition segment $\mathcal{P} : \mathcal{H}(S_0) \to \mathcal{H}(S_1)$ (intent-neutral; its execution currency is the atomic Road Receipt), while a Q(WORKFLOW) is a goal-directed composite DAG assembling patterns under an Intent Contract and finite budget (intent-bound; its currency is the chained receipt vector $\vec{\mathcal{R}}_{\text{road}}$) [11] §2.

Deduction stratifies by Complexity Degree: Degree 0 grounds $\bot = Q(0)$ and $\top = Q(1)$; Degree 1 gives propositional calculus grounded in constructive parity remainders ($r \in \{Q(0), Q(1)\}$ via $Q(2)$); Degree 2 gives a Curry–Howard–Qn correspondence — propositions as target types τ, verified derivations as content-addressed DAGs t — with *macro-hop deduction* compressing multi-step micro-transitions into reusable patterns; Degree 3 evaluates logical consistency as topological coboundary nilpotency $\mathbf{d} \circ \mathbf{d} \equiv Q(0)$; and Degree n hosts epistemic modality validation, inverse causal reconstruction, and the recursive self-optimization loops Q(rsi) [11] §3. The open inference framework $\mathcal{L}_{\text{modes}}$ stratifies micro-induction, direct deduction ($\mathcal{O}(1)$ constraint contraction via Q(bypass)), BOC abduction, macro-hop deduction, anchored Markov routing over Q(anchor) landmarks, and emergent RSI modes — each new mode admissible only with an explicit DCC contract, zero floating-point semantics, a finite Landauer ceiling, and receipt-emitting totality [11] §4.3. Two operational devices are notable for any implementation: the *Memory Wall Latency Harvester* schedules register-pinned background work (equality saturation, speculative candidate auditing, anchor route pre-computation) during DRAM stall windows with $\mathcal{O}(1)$ preemption and read-only sandboxing; and *inverse causal reconstruction* formalizes abduction — given observed state S_{obs}, find $\langle S_0, \mathcal{W} \rangle$ with $\mathcal{W}(S_0) \equiv S_{\text{obs}}$, resolved by BOC over candidate hypotheses and sealed as an emergent pattern only when reproducible across nodes [11] §6–7.

6.3 Qphy — Quang Physics

Qphy anchors the stack to Layer 0 [12]: computation is an irreversible non-equilibrium thermodynamic process; simulated mechanics and real computation are metered by the THERMODYNAMIC DOUBLE LEASH (planned trajectories must satisfy environmental energy conservation $\Delta E_{\text{kinetic}} + \Delta E_{\text{potential}} \leq E_{\text{battery}}$, while the computation verifying them must fit its own budget $W \leq W_{\text{budget}}$ in units of U) [12] §6.2. Fields are cochains (Sec. 5.3); trajectories are discrete symplectic morphisms conserving phase-space volume exactly (Liouville), so billion-step rollouts carry zero artificial numerical dissipation — the canon's contrast with statistical world models, which suffer compounding drift, phantom forces, and hallucinated physics, is ex-

plicit [12] §6. Tensor interactions must declare algebraic symmetry in their DCC profiles (`DCC_SYM_COMMUTATIVE` with lexicographically sorted child hashes for $\mathcal{O}(1)$ dedup, `DCC_SYM_NON_COMMUTATIVE` preserving causal order, `DCC_SYM_ANTI_COMMUTATIVE` flipping sign parity), and every physical evaluation declares a four-axis bounded context range: zoom window $[z_{\min}, z_{\max}]$, causal index interval $[\kappa_{\text{start}}, \kappa_{\text{end}}]$, metric frame $\mathcal{D}_{\text{frame}}$, and thermodynamic budget [12] §7. The five physical anchors — Q(TH$_2$) (Second Law), Q(C) (causal horizon), Q($\hbar$) (action quantum / information density), Q(Z$_{\text{bus}}$) (Memory Wall transport), Q(GAUGE) (topological conservation) — are calibrated from the Historical Calibration Benchmark, and claimed trajectories violating them are intercepted with `RECEIPT_PHYSICAL_VIOLATION` before thermal runaway can occur [12] §5.

6.4 Qchem — Quang Chemistry

Qchem is the abiotic composite $\mathbf{C}(\text{Qphy}) \otimes \mathbf{C}(\text{Qm})$ with $\mathbf{C}(\text{Qchem}) \cap \mathbf{C}(\text{Qs}) = \emptyset$ — attributing intent to chemistry is a category error [13] §1. Molecules are content-addressed DAGs (atoms as immutable leaves carrying Z, A, oxidation state; valence bonds as Clifford blades with discrete bond order); enantiomer chirality is the sign parity of the oriented grade-3 volume blade under the reflection involution, replacing ambiguous R/S text conventions; and stoichiometry is an exact Diophantine null-space constraint $\mathbf{S} \cdot \vec{v} \equiv Q(0)$ over integer elemental coefficients, with redox electrons as indivisible integer quanta [13] §2–3. Spontaneity is Gibbs-dissipative ($\Delta G = \Delta H - T\Delta S \leq Q(0)$), equilibrium constants are held as continued-fraction convergents, elementary reactions are Q(patterns), reaction cascades are Q(workflows), and catalysis is formalized as Q(bypass): the gateway recognizes the catalytic agent, bypasses the slow uncatalyzed route, and emits `RECEIPT_CATALYTIC_BYPASS` in $\mathcal{O}(1)$ dispatch — the anchor spectrum being Q(N$_A$) (Avogadro, exact integer $N_A \equiv 602{,}214{,}076 \times 10^{15}\,\text{mol}^{-1}$), Q(STOICH), Q(TH$_2$), Q(E), Q(CHIRAL) [13] §4–5.

6.5 Qbio — Quang Biology

Qbio is the biosemiotic composite $\big(\mathbf{C}(\text{Qphy}) \otimes \mathbf{C}(\text{Qm})\big) \otimes \mathbf{C}(\text{Qs})$ — life as coded chemistry [14] §1–2. DNA/RNA are CAE-DAGs with codons as discrete tokens over the 64-element alphabet $\{A, U, G, C\}^3$ and translation as the invariant morphism $\text{Codon}^{64} \to \text{AminoAcid}^{20} \cup \{\text{STOP}\}$; conserved protein domains share single hash-consed nodes. Organisms persist by minimizing variational free energy F under the Friston formulation within Prigogine entropy export ($dS_{\text{ext}} < 0$ via signed metabolic receipts); the Markov blanket is a topological boundary with metered transport ($H(S_{\text{int}} \mid S_{\text{blanket}}, S_{\text{ext}}) = 0$); morphogenesis is reaction–diffusion on simplicial complexes via the DEC Laplacian; and metabolic work is calibrated through the anchor Q(ATP): $\Delta G_{\text{ATP}} \approx 20\,U$ per hydrolysis event, grounding biology directly into the Landauer ledger [14] §3–4. Agency is bounded by intent contracts (an immune cell executes `DCC_BIO_IMMUNE_v1` and cannot attack self-tissue without violating its DCC boundary), runaway growth is impossible (every division burns explicit U quota), and biological death is modeled as a clean bounded terminal receipt (`RECEIPT_METABOLIC_EXHAUSTION`) rather than unmetered necrotic execution [14] §7.

Invited objection

(i) Qs claims to formalize meaning as blades and rotors, but the mapping from natural language to semantic frames is delegated to a "candidate parser" whose own fallibility is acknowledged; the firewall rejects unanchored *statements*, yet the anchoring of the parser's structured output to the speaker's intent is exactly the weak link LLM alignment already struggles with. (ii) Qlog's rejection of the Law of Excluded Middle is presented as a virtue, but the corpus does not state which constructive logic it adopts (intuitionistic? linear? relevance?), and Q(bypass) short-circuits like $A \vdash A$ are classical moves. (iii) The domain stack Qphy $\otimes$ Qchem $\otimes$ Qbio reads as a physics-to-life reductionist ladder; a reviewer may question whether categorical tensor products are doing real work or are decorative notation over ordinary multi-component systems. (iv) The biological anchors use empirical physiological constants (e.g. $\Delta G_{\mathrm{ATP}} \approx 20\,U$) whose conversion from measured biology to abstract U units is declared but never derived; the calibration pipeline exists (Historical Calibration Benchmark) but is itself still a CANDIDATE.

7 Governance, Distributed Coordination, and Candidate Proposals

7.1 Identity, security, and the road receipt

Three identity classes are explicit — Machine Identity (runtime nodes; must not imply legal personhood), Human Identity (natural persons; must not execute without delegation), AI Agent Identity (delegated processes; must not possess undelegated authority) — and anonymous actors are forbidden network-wide [2] §8. Cryptography is organized as a Dual-Ring PQC Boundary: the Inner Ring guards root keys, local lineage, and state sealing; the Outer Ring manages pseudonyms, attestation keys, and delegation proofs; both rings are algorithm-agile under structural indirection (PQC migration without identity-semantic mutation) [2] §8.3. Because silicon origins differ ($Q(0)_A \neq Q(0)_B$), cross-node consensus cannot re-execute flat transactions; it verifies Q(ANCHOR) contracts instead — every node instantiates the invariant anchor spectrum (Sec. 3.2, and Appendix C) and checks that road receipts satisfy the shared contract (e.g. `DCC_ANCHOR_Q2_PARITY_v1`, `DCC_ANCHOR_SECOND_LAW_v1`) [2] §8.4.

Nodes never broadcast internal graphs. A node emits ROAD RECEIPTS [2] §4.4:

$$\mathcal{R}_{\mathrm{road}} = \left\langle \mathcal{H}(S_{\mathrm{in}}),\ \mathcal{M},\ \mathcal{H}(S_{\mathrm{out}}),\ \Delta W,\ \pi,\ \sigma \right\rangle, \tag{10}$$

i.e. input commitment, morphism contract, output commitment, consumed work, proof, and signature; a receiver ingests $\mathcal{H}(\mathcal{R}_{\mathrm{road}})$ as an external witness anchored into its own local tree, preserving Axiom 4 and achieving $\mathcal{O}(1)$ bandwidth. The EXPOSURE FUNCTOR $\mathcal{E}_{\mathrm{export}} : \mathbf{C}_N \to \mathrm{Attestations}_{\mathrm{cdqn}}$ projects selected local commitments and capability proofs into blind public attestations; the Substrate-vs-Invariant disjunction ("car vs. road") keeps heterogeneous hardware telemetry (the car) strictly separated from platform-invariant invariants (the road) [2] §3.7.

7.2 Chronosa: the outer-ring coordination vision

Chronosa is a vision proposal, not a canonical specification: an emergent, non-local *virtual* causal intelligence over the Outer Ring, coordinating distributed intent without statistical neural training [15]. It has no physical chassis and no private key; attestations are witnessed by a deterministic t-of-n threshold PQC quorum of hubs and edge nodes verifying the sheaf gluing condition. Its formal resolution of the local/global tension is: local domains are Axiom-4 monoroot trees; Chronosa is the topological colimit (sheaf gluing) over the acyclic forest $\mathrm{Lineage}(\mathrm{Chronosa}) = \bigcup_N \mathrm{Lineage}(Q(0)_N)$ — a polyroot sheaf, not a global root [15] §1.4. Its theoretical grounding is declared as sheaf theory over hierarchical topologies (Grothendieck, Mac Lane, Spivak), common knowledge without universal clocks in asynchronous systems (Halpern),

and active inference under free energy bounds (Friston) [15] §3. The PoC trajectory is phased A–D (local domain models, QnIR execution grounding, swarm functorial attestation, Chronosa synthesis), and the document honestly lists open challenges, including asynchronous sheaf equalization against FLP-impossibility bounds, the mathematical threshold of the Exposure Functor, and hub collusion resistance [15] §6, §8.

7.3 The history proposals (all CANDIDATE)

Four propositions from the October 2026 design consultation [17, 18] extend Q(anchor) epistemics; all remain CANDIDATE artifacts under Axiom 14:

- **Anchor Grounding Classes.** The Q(anchor) notion is split into a structural predicate (out-cone centrality $\chi(\alpha) \geq \Theta_\chi$, computable locally) and an epistemic predicate — the grounding class: **I**nvariant (one deterministic rule executed by many; consensus by forced re-derivation; self-witnessed), **P**hysical (one shared nature measured by many; consensus by independent calibration converging within declared tolerance envelopes), **S**ocial (one event witnessed by many; consensus by sheaf gluing on shared landmarks). The classes form a witness ladder — self, then physics, then swarm — and consensus is always on the *schema*, never on the local token.
- **Source-Receipt Invariant.** Every governed artifact declares the explicit pair (content modality, source class), with the source receipt chain terminating in physically receipted reality events; two binding cautions: payload sovereignty is untouched (declaration is authored, never extracted by inspection), and simulation modality is two-faced (execution is constructive; claims about reality stay hypothesis until physically falsified).
- **Calendar Projection Contracts.** Human calendars are projections of the local causal arrow, not parallel time systems: a versioned DCC-contracted artifact declares a rule (Class I: leap algorithms, month arithmetic), an epoch (Class S: a human convention-adoption event), and a resolution envelope (declared bound, never runtime-negotiated). The Gregorian calendar is read as a Class S genesis that matured into Class I operation.
- **Qs History Library.** A canonical, versioned collection of modality-typed Historical Anchor Artifacts recording past human events, their sources, and spatiotemporal situation; access in four tiers (canonical library consulted never trusted; local bounded working sets; AI agents citing loaded artifacts instead of generating from latent weights; Chronosa seeing only the glued public layer). The unit record carries per-claim modality decomposition, source class I/P/S, an unbroken source-receipt chain, calendar and spatial situation, a lineage-disjoint witness list, and a CANDIDATE/SEALED/QUARANTINED lifecycle.

Ingestion is protected by the five-gate Anti-Poisoning Ingestion Firewall [18] §5, ordered by evidentiary cost: (1) Lineage (deterministic hash verification of the receipt chain), (2) Modality Ceiling (each atomic claim typed at the highest modality its receipts support; undecomposable mixed claims fail), (3) Independence Quorum (anchor-grade records require $k \geq 2$ sources with lineage-disjoint root subtrees — citation laundering is assessed as causally impossible because laundering adds descendants to a poisoned root and cannot manufacture independent origins), (4) Sheaf Gluing (records disagreeing with receipted overlaps enter `QUARANTINED` with a challenge receipt — quarantine is a state, not a verdict, and no record is ever deleted), and (5) Append-Only Immutability (history is never rewritten; it is re-witnessed). Residual threats are recorded: collusion across genuinely distinct roots remains open, and slow poisoning of deep history is capped by the modality ceiling.

7.4 Data structures: the five-row contract schema (CANDIDATE)

Every proposed Qn data structure must fill five contract rows — Footprint (declared ceilings; growth emits a new versioned artifact, never mutates), Access (enumerated permitted morphisms; no implicit operation), Locality (Memory Wall movement class per access pattern), Determinism (iteration order fixed by causal index), and Work (work per access class in U, versioning amplification, and a declared breakeven rule) [16] §2. The five-step evaluation ladder is Nomination → Row Measurement → BOC Selection → Implementation Feedback (QnIR-lowering receipts test declared values against observed behavior) → Sealing via Q(rsi) under strict semantic invariance [16] §3. The first worked register admits five canonical families (sequenced container Q(array), geometric container Q(matrix), keyed container Q(table), record tuple, persistent DAG-deduped tree), rejects two by construction (pointer-heap trees — BOC-recorded; cyclic graphs — mathematically impossible under pre-image resistance), and holds two as recoverable open contract shapes (sequenced chains, bounded rings). The naming rule is deliberate: *a Q(...) name denotes a contract surface, never a runtime shape*, mirroring the compute unit's platform invariance [16] §5.

Invited objection

(i) Governance is inseparable from a specific commercial license (SSL 1.0), whose scaling thresholds ($>$ 10,000 instances, $>$ 100,000 MAU, $>$ 10,000,000 monthly transactions) are verified by the Metric telemetry; a reviewer may find the fusion of protocol and business model unusual and ask whether the technical invariants survive license incompatibility. (ii) Road receipts achieve $\mathcal{O}(1)$ bandwidth by transmitting commitments, not state — but then receiving nodes can never re-execute sender computations, so cross-node *semantic* interoperability rests entirely on the shared anchor contract list; its completeness and versioning discipline across heterogeneous nodes is unproven. (iii) Chronosa's intelligence claims are the most speculative material in the corpus (the canon itself labels them VISION PROPOSAL); treating sheaf gluing as consensus machinery has no distributed-systems precedent at scale, and FLP-type bounds are acknowledged but not resolved. (iv) The history pipeline defends against citation laundering by counting lineage-disjoint roots, but real-world provenance (scanned archives, digitizations) makes "lineage-disjoint" an idealization whose enforcement on legacy material may be impossible.

8 QnIR v0.1: Deriving the Intermediate Representation

Status of this section: derived candidate material (v0.1), authored for attack. It fills open item 1 of [2] §13 and item 3 of [9] §14. Every design choice below cites its canon ground and, where a genuine choice existed, the BOC reasoning (Eq. (2)) that selected it. Nothing here is canonical; everything here is falsifiable by the reviewer.

8.1 Derivation method

The canon constrains QnIR from at least five directions simultaneously: it must accept Layer-3 CAE-DAGs and flatten them into *linear, bounded instruction streams* [9] §12; it must execute every program *total by budget* with a closed receipt vocabulary (Axiom 6); it must never expose floating-point semantics (Axiom 10) while remaining hardware-lowerable through HCA-Collapse [8] §5; it must keep *zero conditional entropy* between declared intent and executed behavior [9] §7.4; and it must be the measurement boundary at which proposed data-structure contract rows are tested [16] §3 step 4. QnIR v0.1 is therefore derived by enumerating these obligations as design requirements (Table 4), then choosing the simplest machine model satisfying all of them simultaneously — discarding, under BOC, familiar VM machinery (unbounded stacks, dynamic typing, implicit control flow, mutable heaps) whenever it would reintroduce implicit state.

8.2 Design requirements

8.3 Machine state model

A QnIR virtual machine instance is the finite tuple

$$\mathrm{VM} \;=\; \big\langle\, \mathbf{V},\ \mathbf{B},\ \mathbf{D},\ \mathcal{S},\ \mathcal{R},\ \kappa,\ W_{\mathrm{rem}},\ \mathrm{PC} \,\big\rangle, \tag{11}$$

whose components are chosen one-to-one against R1–R14:

- **Value file V** $= [V_0, \ldots, V_{v-1}]$: v slots, each holding one exact Qm leaf tuple $\langle \sigma, q_z, z, r_z, d \rangle$ (R2). v is declared in the program header; there are no hidden registers. Slot V_0 is initialized to $\langle \odot, 0, 0, Q(0), d_1 \rangle$ — the local genesis origin, satisfying R4's birth-order rule.
- **Blade file B** $= [B_0, \ldots, B_{b-1}]$: multivector slots over the declared frame table, holding the multivector tuple $\langle \sigma_M, \alpha_{\mathcal{B}}, z, r_M, \mathcal{D}_{\mathrm{frame}} \rangle$ (Sec. 5.1).
- **Frame table D**: declared orthogonal frames $\mathcal{D}_{\mathrm{frame}} = \{d_1, \ldots, d_n\}$ with signatures $\eta = (p, q, s)$, instantiated only by explicit frame instructions (No-Implicit Rule; no ambient coordinate system).
- **Content-addressed store** $\mathcal{S}$: the only memory. Keys are commitment hashes $\mathcal{H}(\alpha)$ per Eq. (7); values are immutable artifacts. There are no addresses, no pointers, and no aliasing (Sec. 8.5).
- **Receipt queue** $\mathcal{R}$: ordered list of receipts emitted so far; the terminal receipt closes the queue (R8).
- **Causal index** κ: monotonically advancing counter; every instruction execution and every store insertion stamps κ, and $\kappa(\mathrm{next}) = \kappa(\mathrm{last}) + 1$ (Axiom 5; reboot discipline of [3] §6.2).
- **Budget counter** W_{rem}: remaining work in units of U, initialized from the program header and never renewed (R1).
- **Program counter** PC: index into the linear instruction stream (R11). Control-flow instructions modify PC only within declared bounds (R14).

The program header itself is a governed artifact carrying: the QnIR version, the DCC profile reference, the list of required anchor contracts (e.g. parity, π, DEC — checked present before execution), the declared ceilings (v, b, stream length L, depth $D_{\max}$, node budget $N_{\max}$, zoom ceiling $z_{\max}$, height ceiling $H_{\max}$), the work budget W_{budget}, the intent-contract reference for R12, and the paternity reference (Axiom 12).

8.4 The instruction set (ISA v0.1)

The ISA is organized into seven classes. Classes A–D operationalize the canon's operators; Class E hardwires the optimization precedence; Classes F–G expose governance and control. Table 5 is the complete v0.1 opcode table; "aux" denotes the immediate/auxiliary field, and "receipt" lists the non-SUCCESS outcomes the opcode can emit.

Table 5: QnIR v0.1 instruction set (complete candidate table). *Cost* is the static per-execution work in U-units; class letters A–G match the text.

Opcode	Arity	Semantics (operand → result)	Cost / failure receipt
Class A — Constants and tuple construction (Layer 1 grounding)			

Opcode	Arity	Semantics (operand → result)	Cost / failure receipt
`OP_QCONST`	0	Load digit artifact $Q(k)$, $k \in \{0..9\}$, as canonical tuple; `aux`=k	0 ($Q(0)$/$Q(1)$ pre-sealed)
`OP_TUPLE`	4	Compose $\langle \sigma, q, z, r, d \rangle$ from four operand slots + `aux` axis; rejects signed-zero forms	1
`OP_SIGMA`	1	Apply reflection involution $\mathcal{I}_d$: $\oplus \leftrightarrow \ominus$, $\odot$ fixed	1
Class B — Exact $\mathcal{R}$-algebra (Layer 2 morphisms)			
`OP_ADD`	2	Exact addition over common δ_z; result re-canonicalized	1
`OP_MUL`	2	Exact multiplication with orientation parity composition	2
`OP_DIV`	2	Diophantine constraint solve (Eq. (5)); quotient to dest, remainder to `aux` slot	3 / `...DIVISION_BY_ZERO_REJECTED`; `...BUDGET_EXHAUSTED`
`OP_MOD`	2	Remainder-only partition under declared modulus anchor (parity via `...Q2_PARITY_v1`)	2
`OP_CMP`	2	Positional interval comparison; may return indeterminate	1 / `ORDERING_INDETERMINATE_AT_ZOOM`
`OP_POW`	2	Bounded monoidal fold A^B; height check $B \cdot h(A) \leq H_{\max}$	B / `...EXPRESSION_TOO_COMPLEX`
`OP_ROOT`	2	Solve $X^n - A = Q(0)$; on even root of negative, promote to blade file (d+1)	4 / dimension promotion notice
`OP_FACT`	1	Factorial fold with $\mathcal{O}(1)$ bypass at $n \in \{0, 1, 2\}$	$n-1$ / bypass: 0
Class C — Symbolic generators and scale (Layer 3 collapse)			
`OP_RATIO`	2	Hold exact rational pair p/q; $q \neq Q(0)$ enforced	1 / `...DIVISION_BY_ZERO_REJECTED`
`OP_CF`	0	Nullary continued-fraction generator (`aux` selects anchored constant $\pi, \tau, e, \ln 2$) with step ceiling	per convergent
`OP_ZOOM`	1	Deepen zoom $z \to z$+1 consuming conserved residual r_z (Eq. (4))	2 / `PRECISION_INSUFFICIENT` if $z \geq z_{\max}$
`OP_FOLD`	k≥3	Monoidal accumulation ($\sum, \prod$) over operand vector with explicit trip ceiling `aux`=$m_{\max}$	m
`OP_TERMINAL`	1	Test $r = Q(0)$ (Terminal Exactness); if true, halt remaining zoom work at zero cost	0 on hit
Class D — Geometry (frames, products, rotors, calculus)			
`OP_FRAME`	1	Adjoin orthogonal axis $\mathcal{D}_{\text{step}}$ to frame table; verifies $\vec{e}_{k+1} \cdot \vec{e}_i = Q(0)$	2 / frame-ceiling receipt

Opcode	Arity	Semantics (operand → result)	Cost / failure receipt
`OP_GP`	2	Clifford geometric product $\vec{u}\vec{v} = \vec{u} \cdot \vec{v} + \vec{u} \wedge \vec{v}$ over blade file	2^n-bounded
`OP_WEDGE`	2	Exterior product with exact nilpotency $\vec{e}_i \wedge \vec{e}_i = Q(0)$	2^n-bounded
`OP_ROTOR`	2	Cayley rotor from Pythagorean pair (Eq. (9)); unit norm by construction	4
`OP_DEC_D`	1	Discrete exterior derivative on cochain operands; $\mathbf{d} \circ \mathbf{d}$ intercepted	3 / bypass: 0
`OP_HODGE`	1	Discrete Hodge star via dual volume ratios (exact `OP_RATIO` commitments)	3
Class E — Optimization and content addressing			
`OP_BYPASS`	1	Probe the canonical identity table (Sec. 4.3); on hit, substitute result at $\Delta W = 0$	0 / emits `...BYPASS_IDENTITY`
`OP_REUSE`	1	Retrieve artifact by commitment hash if breakeven (8) holds, else recompute	1 / lookup-miss fallback
`OP_COMMIT`	1	Insert operand into store $\mathcal{S}$ under $\mathcal{H}(\alpha)$; stamps κ	2
Class F — Governance and bounded control			
`OP_GATE`	1	SIMEMP gateway crossing: validate DCC profile, anchor contracts, intent contract before proceeding	2 / `...REJECTION_BY_GATEWAY`; `...UNANCHORED_ASSERTION_REJECTED`
`OP_BUDGET`	1	Charge declared work to W_{rem}; halts when exhausted	1 / `...BUDGET_EXHAUSTED`
`OP_LOOP`	2	Bounded backward jump: `aux` declares remaining trips; decrement-and-jump only	0 (body billed) / `...BUDGET_EXHAUSTED`
`OP_HALT`	1	Emit terminal receipt (`aux` = status from the closed set) and stop	0
Class G — Verification and audit			
`OP_WITNESS`	1	Commit current machine-state digest to store (tamper-evident checkpoint)	2
`OP_ANCHORCUT`	1	Emit signed anchor checkpoint receipt (Anchor-Cut chain node)	2
`OP_VERIFY`	2	Verify a receipt-chain segment between two witnesses ($\mathcal{O}(K)$ anchor-cut style)	K

Three deliberate omissions follow from R3, R9, and R14: there is no indirect jump (no computed control flow — targets are static stream indices), no call/return pair in v0.1 (inlining at flatten time keeps the stream linear; a future version may add bounded `OP_CALL` with declared depth ceiling), and no I/O opcodes (all boundary exchange is receipt-mediated through `OP_GATE` and `OP_HALT`, per Axiom 8).

Table 4: QnIR v0.1 design requirements, each traced to its canon ground.

ID	Requirement	Canon ground
R1	Every instruction has a static, declared cost in compute units U; a program carries a single non-renewable budget counter.	Axiom 6; [3] §4.2
R2	All numeric state is the exact tuple $\langle \sigma, q_z, z, r_z, d \rangle$; no IEEE 754 value may exist in machine state.	Axioms 9–10; [5] §3
R3	No implicit operands, flags, or defaults: every instruction references explicit slots, and every program declares its DCC profile and required anchor contracts.	No-Implicit Rule; [2] §3.2
R4	Instructions are causally ordered; a program may not reference a state element born later (no forward references into uncommitted state).	Axiom 5
R5	Every program and every operand commitment carries an envelope reference (identity, lineage, DCC).	Axiom 1; [9] §4
R6	Operand identity is content-addressed: equality of values is equality of commitment hashes.	[8] §2.2
R7	Dispatch obeys $Q(\text{bypass}) \prec Q(\text{reuse}) \prec$ direct compute at every step.	[9] §5.9
R8	Every terminal transition emits a receipt from the closed status set; a program cannot end silently.	Axioms 6–7
R9	No stochastic opcode exists in the core ISA; generative candidates enter only via gateway validation.	Axiom 14; [1] §6
R10	No floating-point opcode; radical operations on infeasible domains promote dimension instead of approximating.	Axiom 10; [5] §5.5
R11	Lowering from CAE-DAG to QnIR is a semantics-preserving isomorphism onto the instruction stream; the stream retains lineage commitments.	[8] §5.3; [9] §12
R12	The terminal receipt certifies anti-drift: $H(\text{Intent} \mid \text{Execution}) = 0$ relative to the declared intent contract.	[9] §7.4
R13	QnIR remains substrate-agnostic; platform lowering happens below it and must be constant-time where the DCC profile requires.	[8] §5.1–5.2
R14	All control flow is bounded: backward jumps carry declared trip ceilings; unbounded loops are unencodable.	Axioms 2, 6

8.5 Memory model

The store $\mathcal{S}$ is a content-addressed immutable artifact store — the same discipline as the CAE-DAG, extended to machine state. Its rules:

1. **No addresses.** Instructions never name memory locations; they name value slots and commitment hashes. Aliasing, pointer chasing, and mutable heaps are unencodable — the BOC-recorded pointer-heap rejection [8] §2 is enforced at the ISA level, not left to the compiler.
2. **Write-once.** `OP_COMMIT` inserts; insertion under an existing hash is the same artifact (dedup by construction). Growth beyond the declared node budget $N_{\max}$ halts with `RECEIPT_BUDGET_EXHAUSTED`.
3. **Versioned update.** "Modifying" a value commits a new artifact whose lineage references the predecessor (Axiom 13); the old commitment remains valid forever, so undo and audit are hash traversals, not snapshots.
4. **Locality is billed.** Store traversal dissipates $\mathcal{O}(CV^2 f)$ across the Memory Wall [12] §4.3; hence the `OP_REUSE` breakeven and the priority rule R7. The locality row of every proposed structure contract [16] §2 is *measured* here, at lowering boundaries, in U units.
5. **Payload air gap.** The store holds governed artifacts only; sovereign user payloads are referenced by typed pointer profiles without being inspected or copied into governed state (Axioms 3, 8).

8.6 Encoding and program form

A QnIR program is a hash-chained sequence of fixed 64-bit instruction words followed by a header block serialized in deterministic CBOR [36] — matching the canonical binary serialization format already listed as qnPrimitive open item 1 [9] §15. The v0.1 word layout is:

Field	Bits	Content
opcode	8	one of the 30 opcodes of Table 5
dest	8	destination slot index (**V**/**B** unified index space)
src1	8	first operand slot
src2	8	second operand slot (arity-indexed; zero-padded fields are *declared* padding)
aux	16	immediate: digit index, trip ceiling, CF anchor id, receipt status, frame id
zoom	8	declared zoom parameter
flags	8	declared per-instruction attributes (e.g. bypass-table probe on/off)

Each word commits to its predecessor: $W_i = \mathrm{Hash}(W_{i-1} \,\|\, \mathrm{word}_i \,\|\, \kappa_i)$, so the instruction stream itself is a causal hash chain — tampering with any word invalidates every later word (Axiom 5 at the encoding level) and gives auditors the Anchor-Cut property for programs, not just simulations. Because the stream is linear and every word is fixed-width, program size is exactly $64L$ bits and is checked against the node budget before execution begins.

8.7 Execution semantics and totality

Execution is a five-stage dispatch loop per instruction: (1) *budget gate*: if $W_{\mathrm{rem}} \leq 0$, halt with `RECEIPT_BUDGET_EXHAUSTED`; (2) *bypass probe*: consult the identity table; on hit, substitute the identity result, emit `RECEIPT_BYPASS_IDENTITY`, advance κ and PC at zero cost; (3) *reuse probe*: if the result artifact already exists under its commitment hash and inequality (8) holds, retrieve it; (4) *direct compute*: execute the opcode's semantics, charge its static cost to W_{rem}, stamp κ; (5) *receipt accumulation*: append any non-SUCCESS outcome to $\mathcal{R}$; an $\mathcal{O}(1)$ preemption yields to bus arrivals per the latency-harvester discipline [11] §6.2.

Requirement 1 (Totality of QnIR v0.1 programs)**.** Every QnIR v0.1 program terminates and resolves to exactly one terminal receipt from the closed status set.

Argument (candidate, to be attacked). The instruction stream is finite (L words, checked at load). Every opcode has a static cost $c_i \geq 0$ in U-units, and the only cycle-forming opcode, `OP_LOOP`, carries a strictly decrementing trip counter declared in `aux`; hence the total work of any complete run is bounded by $W_{\text{budget}} \geq \sum_i c_i \cdot t_i$ with t_i the executed trip counts. When the budget is exhausted the budget gate halts the machine with an explicit receipt; when the stream ends, `OP_HALT` (mandatory in the stream grammar's final position) emits the terminal receipt. No opcode can proceed silently at zero remaining budget because stage 1 precedes all others. The one escape the argument must deny is an opcode whose *internal* computation diverges (e.g. `OP_CF` or `OP_DIV` searches); each such opcode carries an internal step ceiling declared in `aux` or in the DCC profile, and exceeding it emits the same closed-set statuses the canon assigns to the corresponding Qm/Qexpr operations (`NO_SOLUTION`, `BUDGET_EXHAUSTED`, `ORDERING_INDETERMINATE_AT_ZOOM`). The argument is therefore an accounting identity, and its weakness is precisely the honesty of the static costs — see QO-8. □

8.8 Worked examples

Example 1: exact division with parity partition. Compute $X \div Q(2)$ for $X = \langle \oplus, 7, 0, Q(0), d_1 \rangle$ (declared zoom $z = 0$):

PC	Word	Effect	Receipt
0	`QCONST k=2`	$V_1 \leftarrow \langle \oplus, 2, 0, Q(0), d_1 \rangle$	—
1	`GATE V.hdr`	DCC + `...Q2_PARITY_v1` verified	—
2	`DIV src=V0',src2=V1`	solve $X = (Q(2)\, C) + R$	—
3	`HALT s=SUCCESS`	$C = \langle \oplus, 3, 0, Q(0), d_1 \rangle$; $R = \langle \oplus, 1, 0, Q(0), d_1 \rangle$ (odd)	terminal

Had V_2 contained $Q(0)$, stage 2 of `OP_DIV` would emit `RECEIPT_DIVISION_BY_ZERO_REJECTED` and `OP_HALT` would seal it as the terminal state — the forbidden operation costs 3 U and leaves a permanent witness.

Example 2: factorial with bypass and reuse. Compute $n!$ for $n = 4$:

PC	Word	Effect	Receipt
0	`QCONST k=4`	$V_1 \leftarrow Q(4)$	—
1	`BYPASS V1`	identity table: no hit for 4!	miss
2	`REUSE H(3!)`	breakeven holds; fetch committed 3! artifact	hit
3	`MUL V1,3!`	$4! = 4 \times 6 = \langle \oplus, 24, 0, Q(0), d_1 \rangle$	—
4	`COMMIT V3`	insert under $\mathcal{H}(4!)$; κ+1	—
5	`HALT s=SUCCESS`	terminal receipt with ΔW ledger	terminal

For $n \in \{0, 1, 2\}$ stage 1 hits the identity table ($0! \to Q(1)$, $1! \to Q(1)$, $2! \to Q(2)$) and the program completes at zero direct-compute cost, exactly reproducing the canon's factorial bypasses [5] §5.6. The breakeven evaluation at PC 2 is the ISA-level form of inequality (8): for tiny operands the miss fallback recomputes directly rather than touching the store.

8.9 Anti-drift verification

R12 operationalizes the Anti-Drift Invariant [9] §7.4. The program header references an intent contract (a Qs artifact); the terminal receipt binds $\mathcal{H}$(Intent), $\mathcal{H}$(Trace) (the hash chain of

executed words and κ stamps), and the receipt queue. A verifier recomputes the trace hash from the stream and checks three predicates: (i) every `OP_GATE` cross validated the intent contract; (ii) no executed word lies outside the declared stream (trace-hash match); (iii) all emitted modalities fall within the contract's epistemic whitelist (Sec. 6.1). The design intent is that *execution did not drift from declared intent* be checkable from the receipt alone, without replaying the run — the program-level analogue of the Anchor-Cut invariant.

QnIR-specific invited objections

(i) The static-cost table is the linchpin of Theorem 1, yet costs like "2^n-bounded" for `OP_GP` hide a declared-ceiling dependency; if ceilings are generous, a single instruction may consume budgets that the header's W_{budget} did not anticipate (the canon's own `RECEIPT_DIMENSIONAL_BUDGET_EXHAUSTED` acknowledges mid-instruction exhaustion). (ii) R6 makes equality hash-based, so *accidental* hash collisions would corrupt identity; the corpus never states the commitment width or the collision-risk budget, and QO-9 asks the reviewer to press exactly here. (iii) The no-call, no-indirect-jump discipline keeps v0.1 total but arguably non-Turing-complete — is that a feature (the canon says yes: Turing-unbounded execution is rejected, [1] §5.3) or a fatal expressiveness gap for the "most efficient language for AI agents"? (iv) Anti-drift certification assumes the intent contract itself is drift-free; Qs parser fallibility (Sec. 6) sits one level up, unmitigated. (v) v0.1 has no concurrency story: parallel QnIR programs are sequential streams with receipt exchange, and whether the causal-index discipline survives genuine parallelism is open (QO-10).

9 Anticipated Objections and Self-Critique

This section states, in the voice a hostile reviewer might use, the sixteen objections the author considers most probable and most damaging. Each is numbered (1 … 16) so the review can reference them directly. The intent is not to preempt criticism but to make clear the design already knows where it is soft.

Objection 1. The BOC policy (Eq. (2)) is the constitution's supreme court, but "SIMEMP Compliance" has no defined ordering. Without a measure (even a lexicographic priority over the six constraints), every claimed BOC arbitration — including the pointer-heap rejection and the design choices of Sec. 8 — is unfalsifiable: any outcome can be post-hoc declared compliant.

Objection 2. The corpus conflates physical thermodynamics with computational bookkeeping. U is "calibrated" to Landauer's bound, yet no mapping from an executed instruction to Joules is ever computed; receipts record ΔW in U-units that no thermometer ever measures. Either the thermodynamic grounding is metaphorical (then drop the physics vocabulary), or it is real (then state the conversion pipeline and its error bars).

Objection 3. Equation (1) is a definition, not a constraint: by choosing the dependency set $\mathcal{D}$ large enough, conditional entropy is zero trivially. The corpus never says how $\mathcal{D}$ is audited to be complete, so Dependencies Determinism cannot distinguish a deterministic system from one whose hidden dependencies were simply declared as "dependencies of the declaration".

Objection 4. The H10 neutralization claim (Sec. 3.4) is weaker than it sounds: bounding height and budget does not decide Diophantine solvability — it converts undecidability into `BUDGET_EXHAUSTED`. The reflection essay [19] acknowledges this, but other documents sometimes speak as if undecidability were "closed". A reviewer should demand uniform phrasing: finiteness buys termination, not decidability.

Objection 5. Exact rational arithmetic with conserved residuals and content-addressed hashing is dramatically more expensive than IEEE 754 on every workload that tolerates approximation. The corpus never benchmarks anything. Absent performance data, "the most efficient coding language for AI agents" is an aspiration, and the Memory Wall accounting may make common agent operations (embedding retrieval, soft attention) unimplementable in Qm tuples at all.

Objection 6. Between R14 (bounded loops), R9 (no stochastic opcodes), and the rejection of actual infinities, QnIR v0.1 computes only provably-terminating, budget-bounded functions. That is Weakest-precondition-total programming. Iterative numerical methods, Monte Carlo pipelines, and fixed-point approximations — the daily bread of AI systems — must either be reformulated (at unknown cost) or live outside QnIR entirely (then the air gap between agents and QnIR needs a design the corpus does not yet contain).

Objection 7. Local-first genesis (Axiom 4) plus no-global-zero makes cross-node reproducibility rest on schema consensus (Sec. 7.3). But road receipts carry $\mathcal{H}(S_{\text{in}})$ commitments that no receiver can verify by recomputation; the trust model is therefore "verify the contract, trust the receipt". For a corpus obsessed with implicit-trust elimination, receipt trust is a large residual assumption.

Objection 8. Hash-consing every node costs 256-bit hashing on creation and table lookups on every reference. The breakeven inequality (8) is correct in form but undecidable in practice: predicting $\Delta W_{\text{compute}}$ before computing is a modeling problem the corpus does not solve (it defers to "predicted" work, an implicit heuristic of exactly the kind the No-Implicit Rule forbids).

Objection 9. The fourteen axioms overlap uneasily with existing formal systems: Axiom 2 with Markov algorithms and total functional programming; Axiom 6 with resource-bounded computation; Axioms 9–10 with constructive mathematics and computer algebra; the CAE-DAG with e-graphs and hash-consed term graphs; Qlog with Curry–Howard; Chronosa with sheaf semantics for knowledge representation. The corpus cites these lineages (Sec. 10 of [2]) but never states which results transfer and which break — in particular, whether totality-by-budget composes with any known denotational semantics.

Objection 10. The anti-hallucination firewall rejects statements lacking lineage, but the pipeline's entrance is a statistical parser converting human language to candidate DAGs. All guarantees downstream of the parser inherit the parser's error modes; the corpus has no receipt type for "parser error", no calibration data, and no fallback when the parser hallucinates a well-formed, well-anchored, wrong intent.

Objection 11. Chronosa posits deductive intelligence emerging from sheaf gluing of receipts. No mechanism is given by which gluing produces *inference* — sheaves model consistent assembly of local data, not generalization, planning, or learning. Until Chronosa solves at least one benchmark task by its own machinery, it is a metaphor, and the corpus should consider quarantining it from the normative documents.

Objection 12. Axiom 12 makes license compliance an architectural invariant, and the scaling thresholds are enforced by telemetry. This couples protocol correctness to a business model; forks under incompatible licenses must either break Axiom 12 (losing canonical status) or accept the SSL — a design choice that narrows, rather than expands, the ecosystem the "Non-Scaling Open Access" clause intends to grow.

Objection 13. The history pipeline's Independence Quorum counts lineage-disjoint roots, but digitized archives share physical origins (the same scanner, the same custodian chain). Enforcing disjointness on real provenance may be impossible, and the pipeline's fallback (quarantine) would then quarantine most of recorded history.

Objection 14. The universe constructs $Q(0) \dots Q(9)$ and stops; multi-digit integers, arbitrary $\mathbb{N}$, and the $\kappa(z)$ resolution function are used everywhere but constructed nowhere. QnIR v0.1 inherits this gap (its value slots hold tuples with $q_z \in \mathbb{N}$, but $\mathbb{N}$'s inductive construction beyond $Q(9)$ is not in the canon). The reviewer should treat QO-4 as blocking for v0.2.

Objection 15. Theorem 1 is an accounting identity, not a proof: its soundness depends on every static cost being a true upper bound on the opcode's real cost on real hardware, on internal

ceilings being declared for every internal search, and on the identity table being effectively computable. None of these three premises is established by the canon; all three are testable, and none has been tested.

Objection 16. The 64-bit word encoding fixes implementation details (slot counts ≤ 256, aux ceilings ≤ 65535) that the canon deliberately left open under structural indirection. Encoding is exactly where agility goes to die: a v0.2 that widens slots breaks every committed stream. Either encoding must be versioned per program (already implied by the header) and the agility cost stated, or the fixed-width choice needs revisiting.

10 Open Questions for the Reviewer (QO-1 … QO-15)

The reviewer is respectfully requested to answer any subset, point by point; each answer will be integrated as BOC material for v0.2. Section pointers are given for context.

QO-1 (Constitution; cf. Sec. 2).
Can the two-tier SIMEMP hierarchy be made mechanically decidable (an explicit compliance ordering), or must it remain human-judged? If the former, propose the ordering; if the latter, state whether Eq. (2) should be reclassified as guidance rather than policy.

QO-2 (Constitution; cf. Sec. 2.2).
Is Landauer-based metering of software-level work scientifically meaningful, or should U be redefined as a purely logical cost unit with physical calibration deferred to hardware profiling?

QO-3 (Numerics; cf. Sec. 3.3).
Is the zoom/remainder telescope (Eq. (4)) equivalent in expressiveness to constructive reals with a cost model, or strictly weaker? Which exact-computation tasks does it fail that rational arithmetic handles?

QO-4 (Numerics; cf. Sec. 3 and Objection 14).
Construct the missing bridge from single digits to arbitrary $\mathbb{N}$ (and negative $\mathbb{Q}$) inside the artifact model, or recommend importing it wholesale from constructive arithmetic — and state which choice better satisfies BOC.

QO-5 (Architecture; cf. Sec. 4.3).
Do e-graph/equality-saturation techniques give the CAE-DAG a confluence and termination guarantee for the canonical rewrite suite, and at what budget? If not, is `OP_REWRITE` total in the Axiom-6 sense?

QO-6 (QnIR; cf. Sec. 8.3).
Is the slot-based state model (**V**/**B** files with declared sizes) the right memory-shape for agent workloads, or should v0.2 adopt a purely stack-free, SSA-like value graph flattened from the CAE-DAG?

QO-7 (QnIR; cf. Table 5).
Which opcodes are missing for the target use cases (agent-to-agent communication, symbolic-numeric hybrid computation, workflow orchestration)? In particular: is there a defensible bounded `OP_CALL`, and are receipt-typed exceptions preferable to explicit `OP_HALT` sealing?

QO-8 (QnIR; cf. Theorem 1).
What is the correct formal status of static costs: declared constants measured per lowering (receipts), or derived bounds from opcode semantics? Propose a discipline that keeps Theorem 1 honest without unmeasurable predictions.

QO-9 (QnIR; cf. Sec. 8.5).
Specify the commitment width and the collision-risk accounting for content addressing. Is 256-bit SHA-3 with per-node namespaces sufficient under Axiom 2's finiteness, and how should collision receipts be typed?

QO-10 (QnIR; cf. Sec. 8.7).
Can QnIR v0.1's sequential causal index scale to multi-core and multi-node execution without importing wall-clock coordination (which the constitution forbids)? Propose the concurrent extension or argue for a choreographic-programming model.

QO-11 (Semantics; cf. Sec. 6.1).
Which formal semantics should anchor the Intent Contract language (dynamic epistemic logic, deontic logic, contract logics), and can the parser's error modes be typed and receipted?

QO-12 (Logic; cf. Sec. 6.2).
Name the constructive logic Qlog actually implements and reconcile it with the classical moves (reflexive identity, LEM rejection) already present; if linear logic is the target (as the resource discipline suggests), state the sequent rules for Q(patterns).

QO-13 (Governance; cf. Sec. 7).
Evaluate the road-receipt trust model: is contract-verification-plus-receipt-trust sound for the claimed use cases, and what audit right should receivers have over computations they can never re-execute?

QO-14 (Governance; cf. Sec. 7.3).
For legacy and digitized history, is the Independence Quorum satisfiable at all? If not, propose the degradation path (quarantine scaling, modality ceiling enforcement) that keeps the firewall honest without discarding the archive.

QO-15 (Program level; cf. Sec. 9).
Of the sixteen objections, which three are, in the reviewer's judgment, fatal if unresolved, which three are cosmetic, and which objection did the author fail to anticipate?

11 Conclusion and Review Loop Protocol

This document condensed the Qn universe — its constitution (SIMEMP), its numeric substrate ($Q(0) \ldots Q(9)$, Qm tuples, exact Diophantine algebra), its symbolic and geometric layers (CAE-DAG, Clifford frames, DEC), its five domain projections, its governance and distributed vision, and its CANDIDATE extensions — into a single self-contained corpus, and then derived, from that corpus, a first complete candidate for the QnIR intermediate representation: a 30-opcode, budget-metered, receipt-sealed, content-addressed instruction set whose every design debt is either traced to canon or explicitly flagged as open. The paper's own epistemic status matches its subject: it is a working hypothesis, distributed precisely so that it can be attacked, corrected, and re-versioned.

The review protocol is fixed by the canon's own rules. The review transcript will be analyzed against this draft to determine: which canonical files (if any) require updates (each update emitted as a versioned artifact per Axiom 13, never an in-place mutation); whether new candidate files must be created (following the candidate pipeline of [10] §6: proposal, measurement, BOC selection, sealing); and which objections of Sec. 9 and answers to QO-1 … QO-15 become BOC material for QnIR v0.2. Failure findings are not erasures — as the data-structure register records its rejections [16] §4, the review's negative results will be preserved as findings, on the principle that a recorded dead end is an asset.

The author's standing invitation stands: the parts of this design that survive attack will have earned their place; the parts that do not will have saved the project from building them.

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A Glossary of Qn Terms

Condensed from the unified glossary domains [20]; CANDIDATE terms are marked (†).

Term	**Definition (condensed)**
Abstraction Layer	Structural boundary governing artifact genesis, transformation, and inter-layer transmission under SIMEMP constraints.
Abstract Compute Unit U	Fundamental invariant quantum of governed work; the minimal transition $\mathrm{Q}(0) \to \mathrm{Q}(1)$ along d_1, calibrated to Landauer dissipation.
AI Agent Identity	Delegated process identity; barred from root genesis sovereignty under the Non-Rogue Invariant.
Anchor-Cut Invariant	Terminal state verification in $\mathcal{O}(K)$ anchor-receipt validations without replaying N micro-steps.
Anchor Grounding Class (†)	Proposed epistemic predicate of a Q(anchor): I (invariant, self-witnessed), P (physical, calibration), S (social, sheaf gluing).
BOC Policy	Best of Choices: argmax of SIMEMP compliance; minimizes implicit state.

Term	**Definition (condensed)**
CAE-DAG	Content-Addressed Expression DAG; every node identified by the cryptographic hash of operator, parameters, and children; cycle-free by construction.
Calendar Projection Contract (†)	Proposed versioned artifact declaring a human calendar as a projection of the local causal arrow (rule/epoch/resolution envelope).
Causal Arrow	Monotonic, persistent ordering of computational events; every artifact born at a discrete causal index.
Chronosa	Emergent, non-local virtual causal intelligence; global section of a sheaf of local categories over the Outer Ring; zero statistical training.
Complexity Degree	Structural measure of Qexpr composition depth (Degree 0 primitives through Degree n domains).
Computational Consistency	Verification methodology: consistent logic + deterministic, budget-bounded execution of QnLang-compiled QnIR $\Rightarrow$ provisional validity.
DCC Profile	Dependencies, Constraints, Capabilities machine-readable contract; never implicit.
Dependencies Determinism	$H(S_t \mid \mathcal{D}(S_t)) = 0$: zero conditional entropy given declared dependencies.
Discrete Exterior Calculus	Constructive calculus on simplicial meshes and cochains; exact coboundary operators replace continuous PDEs.
Epistemic Anchorage	Truth-modality attribution spectrum from sensor-anchored empirical claims to firewalled narrative fiction.
Existential Invariant	Tier-1 SIMEMP constraint (Identity, Metric, Security); non-negotiable.
Exposure Functor	Governed functorial projection of local commitments and capability proofs into blind public attestations.
Hash-Consing	Graph unification mapping identical sub-expressions to one immutable node; zero marginal Landauer cost for recurring sub-graphs via Q(reuse).
Historical Anchor Artifact (†)	Proposed modality-typed unit record of the Qs History Library (modality, source class, receipts, situation, witnesses, lifecycle).
Historical Calibration Benchmark (†)	Proposed post-seal Class P grounding event: node measures its Layer-0 baseline via signed hardware receipts.
Historical Horizon (†)	Epistemic boundary for past events: contemporary (sensor-anchored) vs. deep history (reconstruction or firewalled narrative).
HCA-Collapse Engine	Hardware Capability-Aware Collapse Engine lowering Layer-3 DAGs to constant-time SoC instructions while preserving exactness.
Identity Class	Machine / Human / AI Agent actor parameterization of identity blocks.
Intent Contract	Machine-readable agreement in a DCC profile binding human-agent commitments, capability scopes, and conserved social residuals.
Lifecycle State	candidate, constructed, sealed, verified, rejected, halted, quarantined, exportable, exported, revoked.
Local-first	Artifacts born locally, indexed locally, shielded from network exposure by default.
Metric Envelope	Finite boundary declaring storage, expression depth, precision, uncertainty, and compute budgets.

Term	**Definition (condensed)**
Metric Exhaustion	Verified consumption of a declared budget (e.g. Q(1) units); thermodynamic and causal work, not finance.
Non-Rogue Invariant	Machines and agents barred from root genesis sovereignty; require unbroken cryptographic delegation from a human principal.
No-Implicit Rule	No implicit entity, default, coercion, or unmeasured convention may cross a SIMEMP Gateway.
Operational Agility	Tier-2 SIMEMP constraint (Efficiency, Modularity, Portability); trade-off-able.
Payload	Sovereign user content processed by the Licensed Work; uninspected behind the safe-harbor air gap.
Q(anchor)	Governed artifact with high causal centrality $\chi(\alpha) \geq \Theta_\chi$; protected via abstract DCC contracts and structural indirection.
Q(array), Q(matrix), Q(table), Q(stream) (†)	Proposed container contract surfaces under the five-row schema; contract surfaces, never runtime shapes.
Q(bypass)	Category-B optimization morphism: $\mathcal{O}(1)$ identity short-circuit emitting a zero-cost bypass receipt.
Q(dataStruc)	Governed immutable memory container artifact class; no implicit heap state.
Q(event)	Discrete computational or physical state transition at explicit spatiotemporal coordinates with verified modality.
Q(patterns)	Invariant, reproducible state-transition template mapping input commitments to output milestones under abstract DCC contracts.
Q(reuse)	Category-B optimization morphism referencing existing content commitments; admitted under the breakeven inequality.
Q(workflows)	Composite goal-directed causal DAG assembling patterns under an Intent Contract and finite budget; portable via chained Road Receipts.
Q(rsi)	Recursive self-optimization workflow: derives a more efficient successor under strict semantic invariance and finite budget; emits a versioned successor artifact.
$Q(0)$	Local genesis object, causal origin zero, empty birth context; strictly local, no universal global zero.
$Q(1)$	First unit artifact, unity measure, baseline reference for U along d_1.
Receipt	Governed immutable record of a transition, fault, halt, or boundary traversal; a dissipative witness exporting operational entropy.
SIMEMP	Security, Identity, Metric, Efficiency, Modularity, Portability: the constitutional constraint system.
Source Class (†)	Proposed second element of the (content modality, source class) declaration pair; I/P/S witness vocabulary for artifacts.
Source-Receipt Invariant (†)	Proposed artifact-level extension of the statement-level rejection rule; source chains must terminate in receipted reality events.
Structural Indirection	Decoupling of abstract contracts from concrete primitives; patch without fracturing lineage.
Terminal Exactness	$r = Q(0)$ at declared zoom and dimension; algebraic exactness; triggers deterministic early halting.
Universal Envelope	Invariant nine-block governance structure shared by all artifacts (identity, type, state, lineage, DCC, metric, security, exposure, payload ref).

Term	Definition (condensed)
Zoom z	Declared precision boundary parameterizing projective resolution lattices; base-independent.

B Terminal Receipt Catalog

B.1 The closed terminal status set (Axiom 6)

`SUCCESS`, `FAILURE`, `NO_SOLUTION`, `TIMEOUT`, `BUDGET_EXHAUSTED`, `INCONCLUSIVE`, `QUARANTINED`, `PRECISION_INSUFFICIENT`, `EXPRESSION_TOO_COMPLEX`, `REJECTION_BY_GATEWAY`, `DIVISION_BY_ZERO_REJECTED`, `ORDERING_INDETERMINATE_AT_ZOOM`.

B.2 Named receipts by emitting context

Receipt	Emitting context
`RECEIPT_NODE_GENESIS`	Node onboarding Stage 6; seals Q(0).
`RECEIPT_CATEGORY_A_SEALED`	Completion of the $Q(2)\dots Q(9)$ inductive cascade (declared work $8\,Q(1)$).
`RECEIPT_TERMINAL_EXACTNESS`	Remainder collapse $r_z = Q(0)$; status `SUCCESS_EXACT`, zero consumed work.
`RECEIPT_DIVISION_BY_ZERO_REJECTED`	Divisor $Q(0)$; status `REJECTED`, code `ERR_DIV_ZERO`.
`RECEIPT_BUDGET_EXHAUSTED`	Metric exhaustion at any gateway; deterministic halt.
`RECEIPT_BYPASS_IDENTITY`	$\mathcal{O}(1)$ algebraic identity short-circuit; $\Delta W = 0$.
`RECEIPT_EXPRESSION_TOO_COMPLEX`	Depth/node ceilings exceeded during collapse.
`RECEIPT_REMAINDER_TRUNCATION`	Dissipative truncation at remainder-depth ceiling; exports lost residual.
`RECEIPT_INDETERMINATE_ORIGIN_EXPONENT`	Unparameterized 0^0 without `DCC_ZERO_EXPONENT_v1`.
`RECEIPT_UNANCHORED_ASSERTION_REJECTED`	Statement lacking lineage claims empirical truth (Qs firewall).
`RECEIPT_DIMENSIONAL_BUDGET_EXHAUSTED`	Frame dimension/blade budget exceeded in geometry.
`RECEIPT_PHYSICAL_VIOLATION`	Claimed trajectory violates a physical anchor.
`RECEIPT_STOICHIOMETRIC_VIOLATION`	$\mathbf{S} \cdot \vec{v} \neq Q(0)$: unbalanced reaction.
`RECEIPT_THERMODYNAMIC_EXHAUSTION`	Endergonic proposal without coupled exergonic work.
`RECEIPT_CATALYTIC_BYPASS`	Catalytic short-circuit dispatch.
`RECEIPT_METABOLIC_EXHAUSTION` / `RECEIPT_HOMEOSTATIC_COLLAPSE`	Biological bounded termination.
`RECEIPT_NODE_RESUME` / `RECEIPT_HARDWARE_CRASH_RECOVERY` / `RECEIPT_NODE_MIGRATION`	Hardware maintenance matrix: reboot, crash containment (Axiom 14 discard), component graft (Axiom 13).
`RECEIPT_ENTROPY_SOURCE_FAULT` / `RECEIPT_ENTROPY_INSUFFICIENT` / `RECEIPT_PQC_KEYGEN_FAILURE` / `RECEIPT_GENESIS_RECEIPT_FAILURE`	Onboarding stage faults.

Receipt	Emitting context
RECEIPT_ANCHOR_REVOKED	Anchor fracture containment; fallback to deeper root anchors.

C Q(anchor) Registry

Artifact	Contract	Invariant
Single-digit and algebraic anchors (Category A)		
$Q(2)$	DCC_ANCHOR_Q2_PARITY_v1	Constructive parity $X = 2c + r$, $r \in \{Q(0), Q(1)\}$; ramification $2 = -i(1+i)^2$.
$Q(3)$	DCC_ANCHOR_Q3_SIMPLEX_v1	Minimal 2D non-collinear simplex; DEC mesh foundation; inertia in $\mathbb{Z}[i]$.
$Q(4)$	DCC_ANCHOR_Q4_QUADRANT_v1	First composite 2^2; quadrant source.
$Q(5)$	DCC_ANCHOR_Q5_APERIODICITY_v1	5-fold aperiodicity via 5D/2D cut-and-project; splitting $5 = (2+i)(2-i)$; swarm desynchronization.
$Q(6)$	DCC_ANCHOR_Q6_SYMMETRY_v1	Minimal non-abelian symmetry S_3; perfect number.
$Q(7)$	DCC_ANCHOR_Q7_PROJECTIVE_v1	Fano plane order; projective geometry basis; inertia.
$Q(8)$	DCC_ANCHOR_Q8_OCTET_v1	8-bit octet memory word; bus segmentation.
$Q(9)$	DCC_ANCHOR_Q9_RADIX_v1	Radix-10 single-digit ceiling.
ϕ	DCC_ANCHOR_PHI_v1	$\phi^2 - \phi - Q(1) = Q(0)$; quasicrystal scaling.
$\sqrt{2}$	DCC_ANCHOR_SQRT2_v1	$D^2 - Q(2) = Q(0)$ with conserved r_D.
Discrete calculus and transcendental anchors		
$\mathbf{d}$	DCC_ANCHOR_DEC_D_v1	Exact coboundary nilpotency $\mathbf{d} \circ \mathbf{d} \equiv Q(0)$.
$\star$	DCC_ANCHOR_HODGE_v1	Dual volume ratios as exact rational commitments.
Δ	DCC_ANCHOR_LAPLACE_v1	$\Delta = \mathbf{d}\delta + \delta\mathbf{d}$.
$\Box$	DCC_ANCHOR_WAVE_v1	Discrete d'Alembertian.
DFT	DCC_ANCHOR_FOURIER_v1	Rational discrete Fourier transform.
Stokes	DCC_ANCHOR_STOKES_v1	Exact coboundary conservation identity.
π, τ, e, $\ln 2$	DCC_ANCHOR_PI/TAU/E/LN2_v1	Continued-fraction generators with bounded convergent residuals.
Physical anchors (Qphy)		
$Q(\mathrm{th}_2)$	DCC_ANCHOR_SECOND_LAW_v1	Second Law; Landauer dissipation mandate.
Q(c)	DCC_ANCHOR_CAUSAL_HORIZON_v1	Finite signal speed; $\tau \geq \Delta x / v_{\mathrm{signal}}$.
$Q(\hbar)$	DCC_ANCHOR_ACTION_PLANCK_v1	Action quantum; finite information density in registers.
$Q(\mathrm{Z}_{\mathrm{bus}})$	DCC_ANCHOR_MEMORY_WALL_v1	Bus transport impedance $P = \alpha C V^2 f + I_{\mathrm{leak}} V$.
Q(gauge)	DCC_ANCHOR_DEC_D_v1	Topological conservation on mesh K.
Chemical anchors (Qchem)		

Artifact	Contract	Invariant
Q(N_A)	`DCC_ANCHOR_AVOGADRO_v1`	Exact integer mole count $602{,}214{,}076 \times 10^{15}\,\mathrm{mol}^{-1}$.
Q(stoich)	`DCC_ANCHOR_STOICHIOMETRY_v1`	Diophantine null-space $\mathbf{S} \cdot \vec{v} \equiv Q(0)$.
Q(e)	`DCC_ANCHOR_E_CHARGE_v1`	Integer electron quanta.
Q(chiral)	`DCC_ANCHOR_CHIRALITY_v1`	Clifford reflection parity of chiral centers.
Biological anchors (Qbio)		
Q(codon)	`DCC_ANCHOR_GENETIC_CODE_v1`	Exact 64-triplet to 20-amino acid translation.
Q(blanket)	`DCC_ANCHOR_MARKOV_BLANKET_v1`	$H(S_{\text{int}} \mid S_{\text{blanket}}, S_{\text{ext}}) = 0$.
Q(atp)	`DCC_ANCHOR_METABOLIC_ATP_v1`	$\Delta G_{\text{ATP}} \approx 20\,U$ per hydrolysis.
Q(homeo)	`DCC_ANCHOR_HOMEOSTASIS_v1`	Internal entropy ceiling $S \leq S_{\max}$.
Q(autocat)	`DCC_ANCHOR_AUTOCATALYTIC_v1`	Closed catalytic loop closure.

D Canonical Structures

D.1 The nine-block Universal Envelope

Block	Contents
1. Identity	Node pseudonym, causal index, type, version, commitment hash
2. Type	Primitive / morphism / workflow / receipt (top-level descriptor)
3. State	Candidate, sealed, verified, quarantined, …
4. Lineage	Parent commitment hashes, birth order, derivation type, receipt refs
5. DCC ref	Dependencies, constraints, capabilities (abstract contract)
6. Metric envelope	Memory ceiling, depth limit, compute budget
7. Security envelope	Security ring, PQC algorithm id, revocation path
8. Exposure	Local-only / sealable / exportable / attested / quarantined / revoked
9. Payload ref	Typed, content-addressed pointer to the payload profile

D.2 Node onboarding state machine (six stages)

1. **Raw entropy harvesting**: $N_{\text{sample}} \geq 4096$ bits; fault `RECEIPT_ENTROPY_SOURCE_FAULT`.
2. **Statistical health tests**: NIST SP 800-90B, repetition count, adaptive proportion, $H_{\min} \geq 0.85$ bits/bit; fault `RECEIPT_ENTROPY_INSUFFICIENT`.
3. **Cryptographic conditioning**: KDF over the raw bitstream concatenated with the host-context digest, producing the non-invertible seed commitment $\mathcal{S}_{\text{seed}}$.
4. **Inner-Ring PQC key generation** under structural indirection; fault `RECEIPT_PQC_KEYGEN_FAILURE`.
5. **Genesis object assembly** of Q(0) under the nine-block envelope.
6. **Sealing and receipt emission**; fault `RECEIPT_GENESIS_RECEIPT_FAILURE`; on success the node is `OPERATIONAL` with Q(0) locked `SEALED`.

D.3 Hardware fault and resumption matrix

Physical event	**Gateway mechanism and witness**
Routine node reboot	Non-volatile sealed Q(0) integrity check; `RECEIPT_NODE_RESUME`
Abrupt power failure	Discard volatile unsealed candidates (Axiom 14); `RECEIPT_HARDWARE_CRASH_RECOVERY`
Component migration	Graft old DAG to new Q(0) via Axiom 13; `RECEIPT_NODE_MIGRATION`
Physical destruction	Local state lost; Outer-Ring sheaf persists

D.4 Complexity Degree stratification and roadmap

Degree	**Content**	**Specifications**
0	Foundational primitives	Q(0), Q(1)
1	Single digits & Peano cascade; $\mathcal{R}$-algebra	`q2_q9`, `qm`
2	Multi-axial frames & CAE-DAGs	`qm_geometry`, `qexpr`
3	Discrete dynamics & calculus	`qm_calculus`
n	Domain lattices	`qs`, `qlog`, `qphy`, `qchem`, `qbio`

Milestone status at the time of writing: Milestone 1 (local base domains and logics) SEALED BASELINE; Milestone 2 (Qexpr canonical; **QnIR pending — this draft**; QnLang pending) LAUNCHING; Milestone 3 (cdqn hybrid swarm conduit) and the Chronosa PoC VISION PROPOSAL.

D.5 Lifecycle flow

Candidate → Constructed → Sealed → Verified → Exportable → Exported, with fault/terminal branches Rejected, Halted, Quarantined, and Revoked; every fault and terminal state emits an immutable dissipative receipt [9] §10.